

Proceedings of International Conference on “ARTIFICIAL INTELLIGENCE TECHNIQUES & SMART COMPUTING”

(ICAITSC2026)



26 - 27 Feb 2026

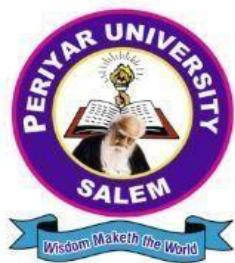
PERIYAR UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

State University - NAAC 'A++' Grade - NIRF Rank 94 State Public University Rank 40
SDG Institutions Rank Band: 11-50, Salem - 636 011, Tamil Nadu, India.



**PROCEEDINGS OF INTERNATIONAL CONFERENCE ON
ARTIFICIAL INTELLIGENCE TECHNIQUES AND SMART COMPUTING
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26.02.2026 – 27.02.2026



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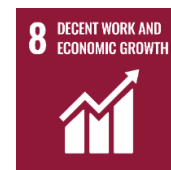
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Salem – 636 011, Tamil Nadu, India



**DEPARTMENT OF
COMPUTER SCIENCE**



**PROCEEDINGS OF INTERNATIONAL CONFERENCE ON
ARTIFICIAL INTELLIGENCE TECHNIQUES & SMART COMPUTING
(ICAITS - 2026)**

Advancing Intelligence for a Smarter Future

26.02.2026 – 27.02.2026



**Organized by
Department of Computer Science**



PERIYAR UNIVERSITY

NAAC A++ - State University – NIRF Rank 94 – State Public University Rank 40

SDG Institutions Rank Band: 11 – 50

Salem – 636 011, Tamil Nadu, India



பெரியார் பல்கலைக்கழகம் PERIYAR UNIVERSITY

State University - NAAC A++ Grade - NIRF Rank 94
State Public University Rank 40- SDG Institutions Rank Band : 11-50
Salem - 636011, Tamil Nadu, India

Dr. V. Raj
Registrar (FAC)

Date : 19.02.2026

PREFACE



In recent years, the landscape of higher education has undergone a significant transformation, driven by rapid advancements in computing technologies and interdisciplinary research. The emphasis on Artificial Intelligence and Smart Computing has become essential in strengthening academic pedagogy and promoting research excellence across institutions worldwide.

I consider it a privilege to appreciate and commend the Convener, organizing committee, faculty members, research scholars, and students for taking the initiative to organize the **International Conference on Artificial Intelligence Techniques and Smart Computing (ICAITS-2026)**. The efforts invested in planning this conference reflect the institution's strong commitment to academic growth and research innovation.

Conferences of this nature provide an excellent platform for scholars and young researchers to present their ideas, exchange knowledge, and engage with experts from diverse domains. Such interactions play a crucial role in nurturing intellectual curiosity and in addressing contemporary challenges through intelligent and sustainable computing solutions.

I am confident that ICAITS-2026 will serve as a catalyst for knowledge dissemination and inspire participants to pursue meaningful research that contributes to national development and global technological progress.

I extend my best wishes to all the delegates and contributors, and I wish the conference a grand success.



[Signature]
19/2/2026
REGISTRAR

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PREFACE

Dr.C.Chandrasekar
Convenor - ICAITSC 2026
Senior Professor & Head
Department of Computer Science



I am delighted to extend my greetings on the occasion of the **International Conference on Artificial Intelligence Techniques and Smart Computing (ICAITSC-2026)**. I sincerely appreciate the Department of Computer Science and the organizing committee for taking the initiative to organize this international conference with great dedication and commitment.

In the present digital era, artificial intelligence and smart computing technologies are playing a vital role in transforming education, research, industry, and societal development. Conferences of this nature provide a meaningful platform for academicians, researchers, and students to deliberate on emerging technologies and share innovative ideas that address real-world challenges.

The conference covers a wide range of contemporary themes that are highly relevant to current and future technological advancements. Such scholarly interactions promote collaborative learning, inspire young minds, and strengthen research culture within academic institutions.

I congratulate the organizing team for their sincere efforts in bringing together distinguished speakers, researchers, and participants from various institutions. I am confident that the conference will result in fruitful discussions and valuable outcomes.

I wish the conference a grand success and extend my best wishes to all the delegates and contributors.

PREFACE

Dr. H. Hannah Inbarani
Conference Chair - ICAITSC 2026
Professor & Head
Department of Computer Science



It is my great pleasure to welcome you to the International Conference on Artificial Intelligence Techniques and Smart Computing (ICAITSC-2026). As Conference Chair, I am honored to bring together researchers, academicians, industry experts, and students on a common platform to share knowledge and innovation.

Artificial Intelligence and Smart Computing are transforming sectors such as healthcare, education, industry, and sustainable development. ICAITSC-2026 aims to foster meaningful discussions on emerging technologies, encourage interdisciplinary collaboration, and bridge the gap between research and real-world applications through keynote sessions and technical presentations.

The conference covers key areas including AI, Machine Learning, Deep Learning, Data Science, IoT, Cyber Security, Cloud and Edge Computing, Computer Vision, and Intelligent Healthcare Systems.

I sincerely thank the organizing committee, reviewers, authors, and delegates for their valuable contributions. I wish all participants a productive and successful conference.

PREFACE

Dr. I. Laurence Aroqiaraj
Organizing Secretary – ICAITSC2026
Professor
Department of Computer Science



It gives me immense pleasure to express my appreciation to the organizing committee for successfully arranging the **International Conference on Artificial Intelligence Techniques and Smart Computing (ICAITSC-2026)**. I thank the Management and higher authorities for their continuous support and encouragement in organizing this academic event.

The field of Computer Science is rapidly evolving with advancements in artificial intelligence, smart computing, and intelligent systems. This conference provides an excellent opportunity for researchers, faculty members, and students to explore recent trends, exchange knowledge, and engage in meaningful technical discussions.

The diverse themes of the conference enable participants to gain insights into cutting-edge research areas and emerging technologies. I am confident that the deliberations and presentations will contribute significantly to academic excellence and research innovation.

I congratulate the organizing team, contributors, and participants for their active involvement and dedication. I wish the conference a great success and hope it serves as a valuable learning experience for all.

ABOUT THE CONFERENCE

The **International Conference on Artificial Intelligence Techniques and Smart Computing (ICAITSC-2026)** serves as a premier multidisciplinary forum that brings together researchers, academicians, industry experts, practitioners, and young innovators from across the globe to exchange knowledge, present cutting-edge research, and explore emerging trends in artificial intelligence and smart computing. The conference is envisioned as a catalyst for innovation, fostering collaboration among diverse disciplines and encouraging research that addresses real- world challenges through intelligent and sustainable technological solutions.

In recent years, Artificial Intelligence (AI) and Smart Computing have transformed the way societies function by enabling data-driven decision making, automation, predictive analytics, and intelligent system design. From healthcare diagnostics and precision agriculture to smart cities and intelligent infrastructure, AI has emerged as a foundational technology shaping the future. ICAITSC-2026 is designed to respond to this rapid evolution by providing a comprehensive platform for discussing theoretical foundations, methodological advancements, and practical applications of AI-driven technologies.

The conference covers a broad spectrum of themes, including but not limited to **AI for healthcare and bioinformatics, medical image analysis, computer vision and intelligent imaging systems, signal processing and edge intelligence, machine learning and computational intelligence, data science, big data analytics, high-performance computing, Internet of Things (IoT), cloud and edge computing, cyber security, and AI- driven solutions for agriculture and sustainable systems**. By embracing both foundational and applied research, ICAITSC-2026 encourages interdisciplinary approaches that bridge the gap between theory and practice.

A key objective of ICAITSC-2026 is to foster innovation through the integration of intelligent algorithms, data-centric methodologies, and scalable computational frameworks. The conference emphasizes research that not only advances scientific knowledge but also contributes to technological progress and societal well-being. Through keynote addresses, invited talks, and peer- reviewed technical paper presentations, participants gain insights into emerging research directions and future challenges in intelligent computing.

ICAITSC-2026 also aims to strengthen **academia–industry collaboration** by providing a platform for industry professionals to share practical insights, emerging requirements, and real-world problem statements. Such interactions help researchers align their work with industrial needs, promote technology transfer, and encourage entrepreneurship and innovation. The conference is particularly beneficial for research scholars and early-career researchers, offering them an opportunity to present their work, receive expert feedback, and build international research networks.

CONFERENCE FOCUS AREAS AND SUSTAINABLE DEVELOPMENT GOALS

ICAITSC-2026 is thoughtfully aligned with key **United Nations Sustainable Development Goals (UN SDGs)**, reinforcing the role of artificial intelligence and smart computing as enablers of sustainable and inclusive development. The conference highlights how AI-driven technologies can be responsibly leveraged to create positive societal impact.

SDG 3 – Good Health and Well-Being

ICAITSC-2026 contributes to good health and well-being by promoting research and innovation in AI-driven healthcare solutions and intelligent medical systems. The conference highlights advancements in intelligent healthcare systems, medical data analytics, predictive diagnostics, computer vision for medical imaging, wearable health monitoring, and telemedicine technologies. By encouraging collaboration between researchers, healthcare professionals, and technology experts, ICAITSC-2026 supports the development of accessible, efficient, and patient-centered healthcare solutions that enhance overall well-being and quality of life.

SDG 4 – Quality Education

ICAITSC-2026 supports quality education by fostering knowledge dissemination, research skill development, and global academic collaboration. The conference encourages AI-enabled learning platforms, intelligent tutoring systems, educational data analytics, and digital learning ecosystems that enhance teaching effectiveness and learner engagement. By bringing together academicians, researchers, and students, the conference nurtures a strong research culture and lifelong learning.

SDG 8 – Decent Work and Economic Growth

By promoting innovation, entrepreneurship, and industry-oriented research, ICAITSC-2026 contributes to economic growth and employment generation. The conference facilitates discussions on AI-driven automation, smart manufacturing, intelligent business analytics, and workforce upskilling. Industry–academia collaboration showcased during the conference supports the development of future-ready skills and sustainable economic progress.

SDG 9 – Industry, Innovation, and Infrastructure

The conference advances research in next-generation intelligent systems, digital infrastructure, smart networks, and scalable AI platforms. Emphasis is placed on innovation-driven solutions that enhance industrial productivity, resilient infrastructure, and sustainable technological development. ICAITSC-2026 encourages research that strengthens national and global innovation ecosystems.

ABOUT THE UNIVERSITY

Periyar University, Salem, Tamil Nadu, is a distinguished public university established by the Government of Tamil Nadu with the objective of providing accessible, inclusive, and quality higher education while fostering advanced research and innovation. The University is named in honor of the renowned social reformer **Periyar E. V. Ramasamy**, whose philosophy of rationalism, social justice, and empowerment continues to guide the institution's vision and mission. Since its establishment, Periyar University has evolved into a vibrant center of academic excellence, committed to nurturing intellectual growth, critical thinking, and societal development.

The University offers a comprehensive range of **Undergraduate, Postgraduate, M.Phil., and Doctoral programs** across disciplines such as Arts, Science, Commerce, Management, Education, Computer Science, and Interdisciplinary Studies. Through its constituent departments and affiliated colleges, Periyar University ensures academic flexibility and diversity while maintaining rigorous academic standards. The curricula are periodically revised to align with emerging global trends, technological advancements, and industry requirements, thereby equipping students with future-ready knowledge and skills.

Research and innovation form the cornerstone of Periyar University's academic ecosystem. The University actively promotes **interdisciplinary research, funded projects, and collaborative initiatives** that address contemporary challenges at local, national, and global levels. Faculty members and research scholars are encouraged to engage in original research, publish in reputed journals, and participate in international conferences. Special emphasis is placed on emerging domains such as **Artificial Intelligence, Data Science, Smart Computing, Life Sciences, Environmental Sustainability, and Healthcare Technologies**, reflecting the University's commitment to impactful and socially relevant research.

Periyar University is supported by robust academic infrastructure designed to facilitate effective teaching and advanced research. The campus houses well-equipped laboratories, advanced computing centers, smart classrooms, and seminar halls with modern audiovisual facilities. The University's central library serves as a knowledge hub, providing access to an extensive collection of books, scholarly journals, e-resources, and digital databases. These facilities empower students and researchers to pursue academic excellence and innovation.

The University places strong emphasis on **quality assurance and academic governance**, ensuring transparency, accountability, and continuous improvement in teaching–learning processes. Through internal quality assurance mechanisms and outcome-based education practices, Periyar University strives to maintain high academic standards and institutional excellence. Faculty development programs, research workshops, and training initiatives are regularly organized to enhance pedagogical effectiveness and research capabilities.

Beyond academics, Periyar University is deeply committed to **holistic student development and social responsibility**. The institution actively promotes ethical values, leadership qualities, and civic engagement through extension activities, community outreach programs, and skill enhancement initiatives. Students are encouraged to participate in social service, environmental conservation, and rural development programs, reinforcing the University’s mission of contributing positively to society.

Periyar University also plays a vital role in fostering innovation and entrepreneurship among students and researchers. Through incubation centers, innovation cells, and research forums, the University encourages creative problem-solving, start-up initiatives, and technology-driven solutions.

With its strong academic foundation, research-oriented environment, modern infrastructure, and commitment to societal progress, **Periyar University continues to emerge as a center of excellence in higher education and research**. The University’s dedication to knowledge creation, innovation, and global engagement makes it an ideal host for international academic events such as the **International Conference on Artificial Intelligence Techniques and Smart Computing (ICAITSC-2026)**. By hosting such conferences, Periyar University reinforces its role in promoting scholarly exchange, advancing research frontiers, and contributing to the global academic community.

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A Layered Explainable AI Framework for Trustworthy Generative Intelligence: The Dynamo AI Approach

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ABSTRACT

The use of large language models (LLMs) in education, research, and decision-making support is growing. However, users are unable to understand how they operate internally, which makes trust and openness difficult to achieve. Current Explainable Artificial Intelligence (XAI) methods usually offer explanations after the fact for models that predict things. These approaches frequently fail to produce content since these systems analyse information using a sequential, probability-driven process. This paper introduces Dynamo AI, an XAI framework with layers that aims to make generative intelligence systems more trustworthy. It does this by putting explainability right into the process of generating content. The framework adopted organizes explainability into four layers: model output, structured ways of showing reasoning, explanations that people can understand, and optional cues for checking information. This study outlines the architecture of Dynamo AI and presents a brief user study that examined users' trust in the system, their satisfaction levels, and the additional time required to receive a response. The outcome of the study was that users trusted and liked Dynamo AI more than standard generative outputs, with only a small increase in response time and incorporating layered explainability offers a practical path toward improving transparency in generative AI systems.

Keywords: Explainable Artificial Intelligence, Generative AI, Large Language Models, Reasoning Traces, Model Transparency, Responsible AI, Human-Centered AI.

I. INTRODUCTION

Large Language Models (LLMs) have demonstrated their effectiveness in tasks such as answering questions, summarizing, and assisting with education (Brown et al., 2020; OpenAI, 2023). Despite the benefits they offer, these systems often behave as opaque black-box models in that people

perceive seamless results but are unaware of how they were produced. This ambiguity raises questions about accountability, trust, and the safe application of these models, particularly in research and education, where fact-checking is essential (Velez and Kim, 2017; Lipton, 2018).

Explainable Artificial Intelligence (XAI) studies stress that being transparent is a must for users to rely on AI properly (Velez and Kim 2017; Samek and Müller 2019). It has been observed that many explainability studies have primarily focused on predictive models, such as classifiers, where most of them demonstrate how input influences output (Ribeiro et al., 2016; Lundberg and Lee, 2017). Independent studies by Wang et al. (2023) and Zhao et al. (2024) have demonstrated that generative models differ significantly from traditional models, as they incorporate outputs sequentially with rational explanations based on the circumstances.

The same researchers, Wang et al. (2023), have additionally researched thought chain models and found these to be more productive and effective when processes include reasoning. But AI proponents warn that these explanations may indicate post hoc rationalizations that may not display the models' internal workings directly. (Zhao et al., 2024; Wiegrefe and Pinter, 2019). Hence, it is postulated that one needs to explain AI more accurately and practically to the operations of the model while maintaining simplicity to utilize it.

II. RELATED WORK

A. Explainable Artificial Intelligence

Explainable Artificial Intelligence (XAI) aims to make machine learning systems easy for people to understand so that they are more transparent, accountable, and trustworthy. Previous research demonstrated that interpretability is dependent on users' expectations and level of knowledge rather than being a single, unambiguous concept (Lipton, 2018). The majority of XAI research for predictive models has concentrated on post-event explanations. By showcasing how the model behaves for individual predictions, methods such as LIME and SHAP provide answers for particular cases (Ribeiro

et al, 2016; Lundberg and Lee, 2017). These methods presuppose a stable relationship between inputs and outputs, although they work best for straightforward data and classification problems. This assumption causes problems for generative models, which create outputs piece by piece and change dynamically with the situation (Samek and Müller, 2019).

Recent surveys highlight that explainability for deep learning systems must increasingly consider system-level transparency rather than isolated prediction explanations (Adadi and Berrada, 2018; Vilone and Longo, 2021). This shift motivates architectures that integrate explainability throughout the model pipeline, rather than relying solely on post-hoc analysis.

B. Explainability in Natural Language Processing

The quick adoption of language models based on transformers has made people more interested in explaining how natural language processing (NLP) systems work (Raffel et al. 2020). At first, people thought attention mechanisms could be explanations. Later research proved that attention weights don't really show how important things are and shouldn't be taken as true explanations (Wiegrefe and Pinter, 2019).

It is a difficult task; like logical thinking, mathematical problems depend on the development of reasoning methods and Chain of Thought (CoT) prompting (Wang et al., 2022; Wang et al., 2023). But the current works give a caution that these reasoning steps may not be a true representation of how the model thinks internally. (Zhao et al., 2024; Narang et al., 2020). To resolve the above issues, later studies have endorsed capturing structured intermediate signs during the generation and mapping them in explanations that are specifically designed for human interpretations (Wiegrefe et al., 2022; Dar et al., 2023). These findings directly inform the structured reasoning layer used in Dynamo AI.

C. Trust and Responsibility in Generative AI

The people's trust and responsibility in AI systems come more from their transparency, consistency, verifiability and accuracy of their outputs (Mitchell et al., 2019 & IEEE 2019). This forms the basis for the design of Dynamo AI. To resolve the above issues, later studies have endorsed capturing structured intermediate signals during the generation and mapping them in explanations that are specifically designed for human interpretations (Wiegrefe et al., 2022; Dar et al., 2023). However, in generative AI, systems often produce fluent responses that appear trustworthy, leading users to

accept incorrect or fictitious outputs without verification (Zhao et al., 2024; Bender et al., 2021).

The solution may rest with responsible AI, which emphasizes uncertainty limits as well as how AI is meant to be utilized (Mitchell et al., 2019; IEEE, 2019; European Commission, High-Level Expert Group, 2020). Things like model cards and system data sheets give structured ways to share these details at a system level, not just for each output (Mitchell et al., 2019).

The current studies advocate that combining explanations with simpler verification cues, such as source attribution or confidence indicators, can refine user calibration and appropriate reliance on generative systems (Liang et al., 2020; Liu et al., 2023). These were incorporated in Dynamo AI to enhance its verification level.

Further, the later works throw caution on Large Language Models (LLMs), which can pose social and cognitive risks when used without transparency, documentation, and governance mechanisms, and thus there is an increased need for trustworthy AI (Bender et al, 2021; European Commission, High-Level Expert Group, 2020).

D. Layered and Human-Centered Explanations

The layered and human-centered explainability researchers contend that the explanations should focus on user goals and cognitive constraints rather than being entirely technical (Miller 2019). These layered explanations offer explorations at multiple levels of detail, which allows users to examine the system behavior incrementally (Brown et al 2020; Liao et al, 2020).

The empirical studies by Kaur et al (2020) and Cheng et al (2022) showed that the layered explanations enhanced user trust and task performance when compared to single-level explanations. It can be noted that the majority of generative AIs do not implement layered explainability, a matter that needs to be further studied by researchers. Dynamo AI has

identified this issue and developed a solution by using layered explanations in the generation pipeline directly.

III. DYNAMO AI FRAMEWORK

The Dynamo AI framework organizes explainability into four layers. Each layer deals with a distinct portion of transparency and trust.

1) Layer 1 – Model Output:

This layer provides the ultimate response utilizing a large language model (LLM). The primary objective in this stage is to ensure the answer is coherent and pertinent to the task, while keeping the underlying details concealed (Brown et al, 2020).

2) Layer 2 – Organized Reasoning Representations in a structured path:

This layer monitors abstract reasoning indicators, including understanding user intent, selecting context, and making retrieval choices. These indicators are maintained in a structured format rather than as unprocessed model actions (Wang et al, 2023)

1) Layer 3 – Explanations in Plain Language:

Structured reasoning details are transformed into easy-to-understand explanations that clarify how the system arrived at its conclusion. These explanations are concise and targeted towards the task, adhering to the principles of human-centered design (Miller, 2019).

1) Layer 4 – Verification and Trust Cues:

Optional cues, like where information came from, how confident the system is, or suggestions on how to check things, are given to help users use the system responsibly and not trust it too much (Lewis et al., 2022).

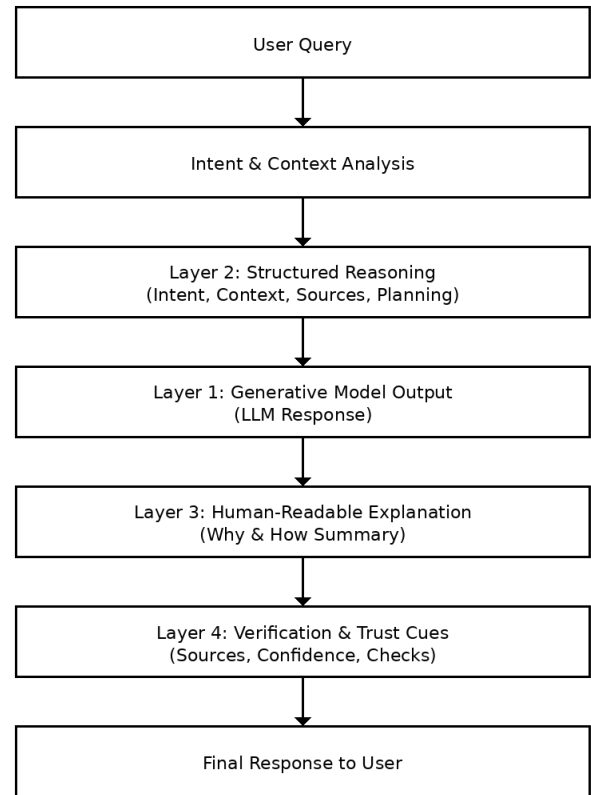


Fig. 1. Layered explainable architecture of Dynamo AI illustrating reasoning, generation, explanation, and verification stages.

IV. SYSTEM ARCHITECTURE

The Dynamo AI system's design makes explainability a top priority while also keeping it modular and scalable, as illustrated in the system architecture below.

A. Frontend Interface

The user interface handles how users interact with the system, including typing in questions, showing answers, and turning explanations on or off. Users can choose to view different explanation layers depending on their needs and expertise, allowing flexible interaction with the system.

B. Generative Core

The core generation part is in charge of creating smooth, context-aware answers using a large language model. Modern transformer-based designs, like GPT-style and T5-style

models, are suitable main builders for this part (Brown et al. 2020; Raffel et al, 2020).

C. Explanation Orchestration Layer

This layer manages recording reasoning steps, turning those steps into explanations people can understand, and adding optional cues for checking information. By maintaining a distinction between generation and explanation, the architecture enables transparency across the system while avoiding inaccurate assertions about directly understanding the model's internal thought process (Zhao et al, 2024).

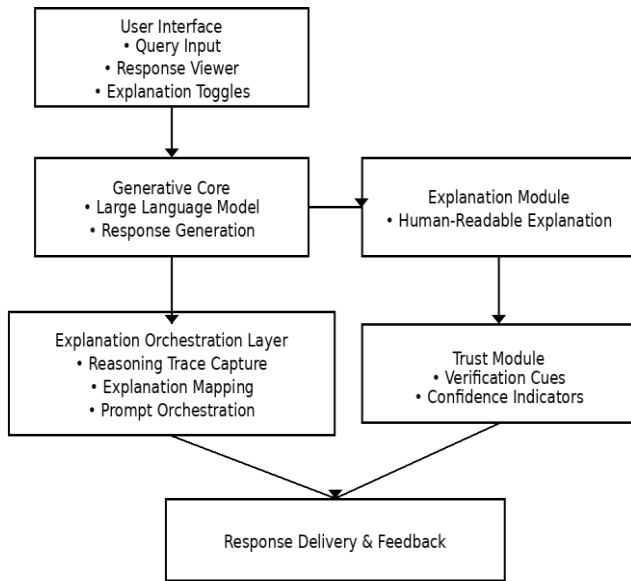


Fig. 2. System architecture of Dynamo AI showing interaction between user interface, generation, explanation, and trust modules.

V. EVALUATION

A small study with 30 people compared standard generative outputs to Dynamo AI output with layered explanations. Trust and satisfaction were measured using surveys with a Likert scale.

TABLE I

PILOT STUDY RESULTS (N = 30)

Metric	Baseline	Dynamo
Perceived Trust (1–5)	3.1	4.0
Verification Satisfaction	3.3	4.1
Relative Response Time (%)	100	109

VI. DISCUSSION AND LIMITATIONS

The study suggests that using layered explainability can make users trust a system more without making it too slow. However, the explanations are based on abstract reasoning signals, not on the full internal states of the model, which are still hidden in large transformer models (Zhao et al, 2024; Wiegrefe and Pinter, 2019).

The current study was validated using a small sample size and future research shall be validated by a larger sample size. Prior research on calibration and trust in AI-assisted decision making suggests that explanation mechanisms can improve appropriate reliance when paired with uncertainty and verification cues (Liang et al., 2020; Liu et al., 2023).

VII. CONCLUSION

This paper introduced Dynamo AI, an explainable AI framework with layers, designed to make generative intelligence trustworthy. By building explainability into the generative process and sorting it into clear layers, the framework helps people use generative AI systems transparently and responsibly. Upcoming efforts will focus on conducting additional evaluations and exploring automated methods for verifying information.

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Intelligent Renewable – Energy Prioritized Home Automation and Load Balancing using Raspberry PI and Predictive

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Abstract - This paper addresses a smart home automation and energy management system that will help to better utilize renewable energy in the daily functioning of the household. The Raspberry Pi used as the central controller to collect real-time data from the current and voltage sensor and is able to control household applications through ION-enabled switching modules. The system is able to better manage a load and balance the use of solar and grid electricity by predictive load forecasting, genetic algorithm and random forest on historical energy consumption, load scheduling, and energy source selection.

Keywords - Energy Management System, Home Automation, IoT, Load Balancing, Predictive AI, Raspberry Pi, Renewable Energy

I. INTRODUCTION

The rising consumption of electricity with the subsequent rise of the use of renewable energy sources have necessitated the need to manage energy in the homes of the contemporary people [1], [2]. The traditional home energy systems tend to be based on the simplistic timers or manual control, which may result in the unnecessary power consumption, higher electricity bills, and the inefficient utilization of renewable resources [3]. Home automation systems are convenient but most of them cannot maximize energy usage in line with the availability of renewable power [4]. The objective of the proposed energy management system is to allocate available energy to household appliances so that renewable energy utilization is done to the fullest degree while ensuring constant supply to critical loads and maintaining a stable system. To minimize the use of the traditional electricity, to decrease the level of carbon emission, and to encourage sustainable living, all home automation systems must give the highest priority to

other renewable energy and still ensure that they do not interrupt important loads [7], [8]. This leads to inefficiency, increased cost and inefficiency in using renewables. The conventional systems are also not able to dynamically balance loads depending on real household energy consumption and supply, resulting in possible over loads or underutilization of renewable energy [10]. The other limitation is that it is impossible to predict or estimate the pattern of energy consumption, and it does not allow scheduling the appliances in advance [11]. These difficulties indicate the necessity of a better system that includes real-time monitoring, smart control, and IoT-based communication to support the improvement of load balancing and focus on the use of renewable energy [12].

II. BACKGROUND

In the case of traditional home automation and energy management systems, the traditional manual control, a basic timer or a predetermined schedule is typically mandatory to control house appliances [4], [9]. Although they are somewhat convenient, the techniques do not offer the opportunity to monitor energy consumption in real-time and prioritize the consumption of renewable energy sources such as solar energy or wind energy [1], [5]. This frequently leads to poor energy use, higher electricity prices and low sustainability [3], [9]. The other disadvantage is the fact that conventional systems cannot effectively balance loads. The allocation of power is not dynamic based on the real household consumption or availability of renewable energy [10], [14]. As a result, it may cause such problems as overloading certain circuits or the inability to use renewable sources maximally [15], [11]. Moreover, the conventional systems are unable to forecast or foresee the domestic energy consumption habits, and thus cannot schedule appliances in advance to run during peak demand times [11].

Table 1. Comparative Analysis of Existing and Proposed Systems

renewable energy [5], [6]. By the intelligent scheduling and control of appliances, homes can maximize solar, wind, or any

Feature	Existing System	Proposed System
Control Mechanism	Manual/timers	IoT-enabled automated control
Renewable Energy Utilization	Limited and non-prioritized	Renewable-first priority-aware allocation
Load Balancing	Static or limited	Predictive, priority-based scheduling
Adaptability to Demand Changes	Low	Improved through short-term demand estimation
Grid Dependency	High	Reduced through renewable prioritization

New developments in smart home energy management systems have continued to involve the application of machine learning and reinforcement learning methods to deal with the issue of demand variability and integration of renewable energy sources [10], [18], [34]. Hybrid IoT-based architectures have been considered to implement adaptive control of solar, wind, and energy storage devices to improve the efficiency of operations and minimize carbon emissions in the residential setting [15], [16], [32]. Other research works have been put on optimization-based methods, such as fuzzy logic and metaheuristic methods to assist real-time load scheduling and better system efficiency [4], [23]. Moreover, a recent systematic literature review (2018-24) of the smart home energy management systems observes the increasing significance of predictive and AI-based decision models, as well as the gaps in full demand response applications that incorporate machine learning, decentralized control, and real-time IoT data [31].

Not only that, many energy management solutions which already exist lack the flexibility and intelligence required to adapt to different changes in energy generation and user behavior [12]. They act according to unchanging settings and do not learn or optimize with reference to past experience, which leads to poor-quality decision making. Their inability to react effectively to the changes in the supply of renewable energy or to changes in demand is hindered by the lack of real-time data analytics, IoT-based communication, and predictive control mechanisms. This not only causes underutilization of renewable energy but also affects the attempts to attain a sustainable and efficient energy ecosystem [16], [17]. These constraints highlight the increased importance of having a more sophisticated system that combines IoT, artificial intelligence, and smart grid technology to manage real-time monitoring, smart load distribution, and optimal usage of renewable sources of energy.

III. SYSTEM DESIGN AND METHODOLOGY

The offered system will combine IoT-based devices with predictive control techniques to prioritize renewable energy and provide a more efficient approach to household loads [4], [10], [15]. It is designed with a focus on the real-time monitoring of energy usage, efficient distribution of available power, and smarter usage of renewable sources [16], [17].

A. System Architecture

The most important component of the suggested system is a Raspberry Pi, which is the central processing unit that coordinates the work of data acquisition, analysis, and control of appliances [23]. As Figure 1 illustrates above, the Raspberry Pi is a hub that interconnects the different sensing and control devices including smart meters, voltage and current sensors, and IoT-enabled switches using wireless communication systems such as Wi-Fi, Zigbee or Bluetooth Low Energy (BLE) [24], [25]. This crucial unit works together with sensor modules to perform predictive analysis, and manage the household appliances through relay modules and smart plugs. The general arrangement of the whole system is shown in the figure below.

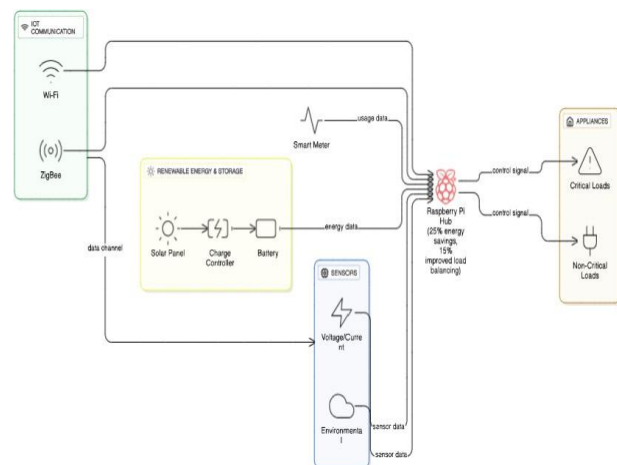


Figure 1. IoT-based home energy system with Raspberry Pi hub, sensors, solar panel, and appliances.

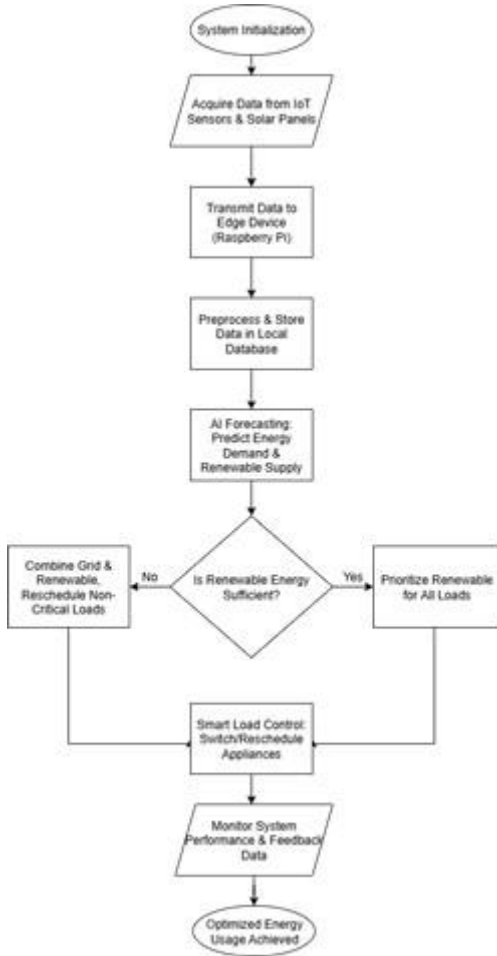


Figure 2. System-level pipeline (from data collection to action)

Decision-making in the suggested framework is structured in multi-stage processes. In the first place, a Priority-Based Scheduling (PBS) mechanism is a tool that classifies household appliances according to preset priority levels and operational constraints into critical, semi-critical, and non-critical loads [3], [9]. Predictive load forecasting is run directly on the Raspberry Pi in a Random Forest regression model, predicting the short time energy demand and renewable availability based on the historical consumption data and the current sensors [6], [29].

$$P_{total}(t) = \sum_{i=1}^N P_i(t)$$

$$\sum(A_i \times L_i) \leq E_{Available}$$

Where several realistic scheduling assignments meet the system constraints, a Genetic algorithm (GA) is used to compute an optimum load assignment, which maximizes the use of renewable energy while maintaining continuous operation of vital loads and the occurrence of unnecessary switching activities is reduced to a minimum [1], [23].

A. Renewable Energy Prioritization

The system suggested is aimed at giving preference to the use of renewable energy to decrease the reliance on the principal power grid. The controller is smart to monitor the generation of solar and wind energy and intelligently decide the amount of available renewable energy at any given time [18], [19]. Important house loads like lighting and refrigeration are first connected so that they can operate continuously even when the energy available is fluctuating. This method does not only enhance the use of renewable energy, but also reduces wastage through effective distribution of power to the necessary appliances [1], [5].

There is also the aspect of using intelligence in shifting or deferring non-essential loads like washing machines or air conditioners when there is insufficient generation in renewable sources to meet the demands of the system [24], [27], [28]. This is a scheduling mechanism that ensures there is a load balance, and the power distribution between renewable sources and grid sources is optimized. The system allows full use of renewable energy and enhanced energy efficiency by combining real time monitoring and predictive control [12], [16], [29].

B. Load Balancing Algorithm

The main goal of the load balancing plan is to achieve the efficient distribution of energy among all the household appliances and give preference to renewable sources without jeopardizing the stability of the system. [18], [22], [30]. Working on discrete time steps (t), the process is controlled in a closed loop fashion where real-time sensing and short-term demand prediction are being used. The real-time current data (t) will be read by the Raspberry Pi controller by using the ACS712 sensor in every control cycle. We can thus estimate the instant power consumption as:

$$P_i(t) = V_{rms} \times I_i(t)$$

Here $P_i(t)$ represents the power demand of the i th appliance, and V_{rms} is the standard supply voltage. The system will calculate the aggregate load at time (t) by summing individual demands [11].

We will classify appliances into 3 groups; critical, semi-critical, and non-critical categories. The scheduling logic allows for a more proper power constraint:



Figure 3. Flowchart of the predictive AI-based load forecasting and scheduling process.

C. Cybersecurity and Data Integrity Considerations

The IoT devices that are linked to the proposed energy management system are identified and authenticated during the exchange of data. To avoid entry of false data and control commands by unauthorized devices into the system, lightweight authentication systems are used which rely on identifiers of the devices and secure tokens [7], [19]. IoT sensors, the Raspberry Pi controller, and user interfaces are expected to communicate via secured channels via standard encryption protocols, including the TLS, to ensure confidentiality and integrity of energy information and prevent the risks of eavesdropping and man-in-the-middle attacks [8], [15]. The edge controller implements role-based access control which limits system configuration and control operations to authorized users only and prevents malicious or accidental changes in the energy scheduling parameters. High-complexity intrusion detection systems and

cryptographic performance analysis are out of the scope of this paper and are cited as the future research directions.

Role-based access control is applied on the edge controller to ensure configuration and control operations of the system are performed only by authorized users to avoid the occurrence of malicious or accidental changes in the energy scheduling parameters. The further development of advanced intrusion detection mechanisms and detailed cryptographic performance analysis falls outside the boundaries of this work and is marked as one possible avenue of research directions in the future.

IV. RESULTS AND DISCUSSION

The system was deployed based on a Raspberry Pi, appliances that had the capability to work with IoT, and predictive control procedures [14], [28]. Its effectiveness was measured in terms of saving energy, load regulation, and comparison with the traditional home automation systems.

Table 2. Appliance Load Distribution Example

Appliance	Load(W)	Peak Status	Hour	Off-Peak Status
Refrigerator	150	Always ON		Always ON
Air Conditioner	1000	ON (if solar > 500W)		Deferred if low solar
Washing Machine	500	Deferred		Scheduled
Lights & Fans	200	ON		ON
EV Charger	1500	Deferred		ON(off-peak)

A. Energy Savings Achieved

The system suggested showed a significant energy-saving based on the intelligent prioritization of renewable energy sources and the use of adaptive appliance scheduling. The controller precisely averaged the production of solar power, demand of the household, and weather patterns to make sure that the renewable energy is used prior to accessing the grid [5], [7], [8]. Such strategy dramatically decreased the amount of redundant grid demand and enhanced the efficiency of locally accessible renewable energy, specifically when the amount of solar is high [1], [21].

Due to these optimization, the system saved energy, as season changed, and solar panels changed their efficiency [27], [28], [31]. This lessening of grid-dependency had a direct effect on reducing the cost of electricity and also reducing the emissions of carbon. In the long-term use, these findings substantiate the prospects of the system to improve the energy sustainability at the home scale and are in accordance with the global energy efficiency targets [24], [22], [30].

B. Load Balancing Efficiency

The system greatly ensured load balance among all appliances that were connected and it was due to the constant monitoring of the real time energy consumption and it reallocated power depending on the level of renewable generation [10], [12], [14]. To evaluate available renewable energy, we will have to combine solar generation output ($P_{solar}(t)$) with the battery's state-of-charge (SoC)[9], [15].

$$E_{available}(t) = P_{solar}(t) + P_{battery}(t)$$

A forecasting module is able to predict short-term load demand ($P_{demand}(t+1)$) by analyzing the historical patterns by including current sensor readings [17], [25], [29]. This step enables the scheduler to predict future needs of energy, and as such, helps in averting possible overloads prior to their occurrence to it.

$$\max F = \alpha \frac{E_{renewable_used}}{E_{total}} - \beta U_{critical} - \gamma S$$

When the predicted demand ($P_{demand}(t+1)$) remains below available energy ($E_{available}(t)$) the system will dynamically schedule both the critical and non-critical loads. However, when the demand ends up surpasses capacity, the system will enter the energy constrained state which operate only on the very important loads and delay non-essential loads [11], [13], [16].

Online training is done at the edge with the deployed model through training offline with historical information. The predictive output is only used to aid the scheduling decisions but not accurate numerical forecasting. This paper will not include detailed accuracy benchmarking.

C. Comparison with Conventional Systems

The proposed framework had great strengths compared to the traditional home automation and energy management systems in terms of flexibility, effectiveness, and sustainability [3], [4], [218]. The current systems are usually based on fixed schedules or manual controls that are unable to react on the changes in energy demand or generation in real-time. The proposed system, on the contrary, makes use of IoT-based sensors and control logic to allow the efficient use of renewable energy and dynamic decision-making [6], [7], [8].

Traditional home energy management tools are generally based on set schedules or manual control measures, which provide a low level of flexibility to dynamic demand patterns and varying renewable energy supply [9], [30]. Newer AI-based solutions, such as deep reinforcement learning and metaheuristic search algorithms, have shown better decision making ability but they may require centralized calculation, large training sets, or complicated system architecture which

can make them less useful with low cost residential implementations [10], [23], [34]. Moreover, the system required reduced reliance on grid electricity, which resulted in greater reliability with outages and a significant decrease in operational expenses [9], [19], [28]. Its improved load scheduling and sharing capabilities enable it to have a higher level of flexibility in controlling energy flow particularly in the conditions of variable renewable supply [5], [25], [31]. All these advancements are a clear demonstration of how the proposed model will be far better than traditional energy management methods in providing intelligent and sustainable energy control that is user-responsive.

In this way, the offered IoT-based energy management system showed significant changes in the energy efficiency and operational stability. The system has successfully minimized reliance on the traditional power grid by focusing on renewable energy sources as well as scheduling the use of appliances intelligently [6], [8], [19]. The load balancing system maintained a consistent power distribution without experiencing over loads and inefficiencies [10], [14]. In general, it was shown that the system was adaptable to different energy production and consumption trends, which facilitated sustainable and affordable domestic energy management [12], [20], [31].

D. Scalability Considerations and System Limitations

The suggested system is developed and tested to work with one residential house, and extensive experimental testing on a large scale in more than one house has not been performed. The system architecture at the appliance level is capable of supporting heterogeneous loads with different levels of priority, and the complexity of scheduling increases linearly with the number of connected devices, which is why it is appropriate in the case of typical household scale deployments.

In a neighborhood scale scenario, the architecture can be conceptually designed by implementing autonomous edge controllers in all households and sharing logs of aggregation of energy rather than random groups of raw data. Such a decentralized structure does not have centralized bottlenecks and maintains local decision autonomy. The proposed priority-aware scheduling logic may be used in the smart microgrid or virtual power plant (VPP) contexts as a local control layer, and more coordination levels control the inter-home energy exchange and grid interactions [22], [27].

The computation of communication overhead, inter-controller coordination, and optimization on a large scale involving a number of households is not present in this work and can be termed as a future research direction.

V. CONCLUSIONS & FUTURE RECOMMENDATIONS

The paper discussed an IoT-based home energy management system that is aimed at giving more priority to the use of renewable energy and enhance the load scheduling in a residential setting. Combining real-time energy monitoring with priority-aware, predictive control at the edge, the proposed structure proves the possibility of eliminating the need to depend on the traditional power grid and still be able to reliably run critical household loads.

The central controller is implemented using a Raspberry Pi, which allows the system to be deployed at a low cost and with adequate flexibility, thus being applicable to small-scale residential applications. The suggested architecture focuses on systematic decision-making, first-generation use of renewable sources, and flexible load control instead of having predetermined schedules or human control.

Future research will involve the expansion of the logic of schedule with the help of such developed optimization algorithms like Particle Swarm Optimization and improved Genetic Algorithms and consider new sensing and predicting options. Moreover, massive validation and extensive performance benchmarking and extrapolation to community-level smart grid or microgrid settings are termed as notable research directions.

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AI-Based Chatbot for Student Skill Assessment, Career Guidance, and Job Recommendation

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Abstract—This research is about designing a smart assistant powered by Artificial Intelligence (AI) that can combine performance prediction, career recommendation, and skill-gap analysis for student decision-making. The suggested approach relies on the multi-modal academic, behavioral, textual, and temporal data, apart from advanced preprocessing, also contains optimized feature selection and a hybrid TGNN-BiLSTM model for precise performance prediction. A career suggestion system based on the Knowledge Graph provides awareness of the career-related context, whereas a Siamese Neural Network skilfully detects the individual skill gap and suggests the perfect learning material. The integrated system is more stable concerning prediction, and the recommendation is relevant, as well as skill-gap identification. In sum, the work asserts that combining multi-format data with sophisticated AI models represents a complete, modifying, and dependable way of support. The framework is significant since it advances aspirants' readiness for the labour market, facilitating the provision of orientation services and delivering personalized educational and career interventions virtually endlessly to a large number of beneficiaries.

Keywords—Personalized guidance, Skill gap analysis, Career pathways, Temporal data, Artificial Intelligence, Career recommendation.

I. INTRODUCTION

In today's world, students face difficulties in identifying their strengths, career interests, and job opportunities [1]. Traditional methods for career guidance often don't work well because they may be outdated, too general, or not available in some places. So, they can't fulfil the student's needs [2]. This system helps to bridge this gap by offering personalized insights, including student skill assessment, career guidance and job recommendation [3]. If the student is planning for a career, it will help to guide them, reduce their confusion and help them to focus on important skills [4][5]. They also provide access to professional mentors, help students learn about career trends and make them aware of opportunities outside their local area [6][7]. With the help of technology, students get real-time information and strong support for both their academic and career growth [8]. Fig 1 shows the basic education system.

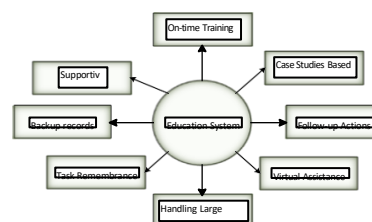


Fig. 1. Education System

The skill assessment is the foundation of this system, a comprehensive evaluation of a student was done consisting of knowledge, technical skills and soft skills such as communication, leadership, and problem-solving [9][10]. The weaknesses were revealed, thus enabling targeted improvement [11]. The assessments provide actionable feedback, and they also recommend the relevant courses or certifications that can guide students in their skill enhancement [12][13]. Goal setting is facilitated, time-tracking of the student's progress is done, and the students are made ready to face the industry's demands and competitive exams [14]. Over a time, system keep checking the student's progress and updates the recommendations, making sure the learning plan stays helpful and suitable for their growth [15].

Existing AI-based app for personalized career guidance, job suggestion, and skill development [16], Smart system matching students with suitable jobs based on interests and skills [17] and AI learning system providing personalized, interactive and visual feedback [20] provide valuable support for student learning and career planning. However, these systems can sometimes rely too much on personal data, which may lead to inaccurate or biased suggestion, and they often cannot fully understand personal or unique situations like a human advisor.

A. Motivation

Finding the right career path and preparing for the future global job market is becoming more complex, and the skills required are changing faster, making it hard for students. Traditional career counselling is often considered insufficient, as it rarely provides personalized guidance is easily accessible or adaptable to new industry trends. Many students are not aware of their strengths, skill gaps, or the career opportunities that suit them best. The AI-Based Chatbot for Student Skill Assessment,

Career Guidance, and Job Recommendation is aimed at helping students know their strengths, overcome skill shortages, and make the career planning process more engaging, reachable, and tolerable by using technologies such as machine learning and natural language processing.

B. The key contribution of the research given in the following,

- A framework designed for an AI system that combines functions, including predicting students' performance, providing career guidance, and assessing gaps in skills.
- The knowledge and joint use of diverse data formats (numbers, text, and time-series) enriched the understanding of the students more than one standalone dataset could bring.
- Various advanced preprocessing methods such as TF-IDF, KNN imputation, normalization, and time-series mapping improved accuracy and quality in several ways.
- Advanced feature extraction techniques were employed that captured the trajectories of student academics, engagement, emotional well-being, and skill level.
- The NSGA-II algorithm was used to select the feature set to ensure maximum accuracy and minimum processing cost.
- TGNN-BiLSTM was used to accurately predict performance; a KGE was used for customized career advice; and SNN was used for skill-gap analysis to authentically engage the learners was the final step.

TABLE I. CHALLENGES IN EXISTING METHODS

Sl. No	Author	Techniques	Advantages	Disadvantages
1.	Bhimavarapu (2025)	NLP (VADER, BERT, SVM, LDA)	Understands student oral-presentation issues	Unclear / varied responses reduce accuracy
2.	Manasa et al. (2025)	ML, Data Mining, Random Forest	Predicts skills, creates learning paths	Sensitive to varied student data
3.	Rahayu & Bablu (2024)	Learner profiling, ML, Adaptive assessments	Personalized learning pathways	Changing skill demands, varied performance
4.	Bahalkar (2024)	Preprocessing, Feature engineering, LSTM	Personalized career suggestions	Missing data, fairness issues
5.	Panthee et al. (2023)	Big Five, VAK, ML	Combines personality + learning style	Self-reported data may be unreliable

6.	Gedrimiene et al. (2024)	TAM + CDM analysis	Understands user needs in AI tools	Limited personalization, narrow options
7.	Guleria & Sood (2023)	ML + XAI (NB, LR, DT, SVM, KNN)	Transparent career guidance	Interpreting black-box models is difficult
8.	Feitosa et al. (2025)	Ontology learning + alignment	More relevant feedback for answers	Ontology building is complex
9.	Omar et al. (2024)	PRISMA, Three-axis model	Clear classification of AI learning methods	Diverse learner needs
10.	Wang (2024)	Blockchain + AI/ML	Secure job recommendations	User engagement + data security issues

Table I shows the challenges in existing methods. The study demonstrates the advancements in AI-driven career guidance, student learning, and skill assessment. However, several gaps remain: existing job and skill recommender systems often lack multiple data sources, personalized adaptability, and alignment with educational needs. Likewise, AI applications in education show limitations in scalability, real-time feedback, and the integration of theory with practice. Motivated by these gaps, there is a need for an adaptive platform that combines personalized recommendations, competency-based assessment and interactive learning. By addressing both educational and professional challenges, such a system can enhance career guidance, skill development, and decision-making.

Organization: The proposed methodology is discussed in Section II and the results acquired are manifested in Section

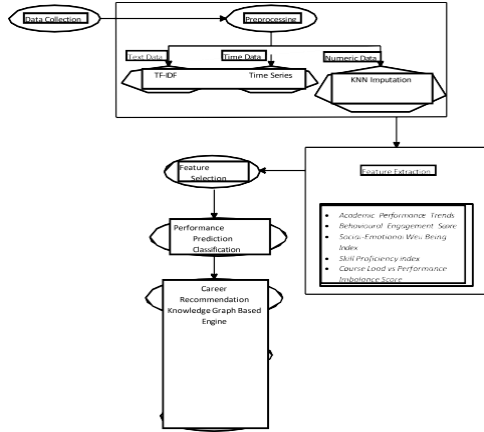
III. This research is concluded in Section IV.

II. METHODOLOGY

The proposed framework employs a multi-stage AI approach employing data pre-processing, feature engineering, predictive modelling and intelligent recommendation approaches to provide personalized student supports. Using multiple inputs including numeric, text and time series-based academic and behavioural data, the data is then cleaned, normalized, and vectorized to achieve high uniformity. Subsequently, the data is scrutinized for significant indicators, including performance trajectories, engagement levels, and proficiency in skills and competencies. By means of an evolutionary feature selection method (NSGA-II) the indicators are filtered and a final subset is evaluated. The temporal relationships in this purified dataset are modelled through a Temporal Graph Neural Network integrated with a BiLSTM (TGNN-BiLSTM) to predict student outcomes with high precision. Based on these forecasted insights, a knowledge graph-based recommendation tool recommends suitable routes to careers. A Siamese Neural Network (SNN) is also employed to examine

skills gaps so as to focus on missing skills and recommend learning opportunities thereby facilitating an engaging AI chatbot environment. Fig 2 shows the proposed methodology framework.

Fig. 2. Framework Architecture



A. Data Acquisition

The system uses a dataset [26] with multiple references containing academic, behavioural, temporal, and skilled characteristics of students by exploring three modalities:

1) Numeric data

Includes records of GPA, attendance percentages, scores obtained on assignments, scores obtained on quizzes and tests, number of views of a page, and activity history. These quantitative features offer an objective snapshot of a student's academic standing and engagement.

2) Textual data

Includes teacher feedback, peer feedback, open-ended responses, and chats. These narratives often indicate the emotional state of the students, their motivation, comprehension challenges, and behavioural passivity. replacing missing values with standard mean values can change variances in equation (3).

3) Temporal data

Has longitudinal capabilities, including weekly GPA growth, term progress through the course, shifts in engagement over time, and learning routine tendencies. Being able to capture the fluctuating effects of time on learning is very important to be able to illustrate learning trajectories and the first indicators of academic decline.

B. Pre-processing

The preprocessing phase consists of the cleaning,

transformation, and preparation of raw data to be suitable for the models. Preprocessing correct errors, noise, missing values, and inconsistencies before any analysis takes place.

It increases model accuracy and stability, and thereby increases model performance by standardizing the different types of data.

1) Text Data Preprocessing

TF-IDF (Term Frequency–Inverse Document Frequency) is a method of transforming the text into numerical vectors. It provides a quantification of the importance of a word in a document as opposed to all the documents in the entire dataset. TF-IDF is an appropriate method because it will take the weight down on many common words, while highlighting word terms that might have meaning regarding a student's feedback or comments using equation (1). TF-IDF generates lightweight interpretable features, and generally outperforms simple counts, because it does provide some importance focus on words consider to be informative.

$$Tf - (a, b) = Tf(a, b) \times Idf(a) \quad (1)$$

Where,

$$Tf(a, b) = \frac{\text{Frequency of } a}{\text{Total words in } b}$$

$$Idf(a) = \ln \left(\frac{N}{1 + n_a} \right)$$

a refers word, b refers document.

2) Time-Series Data Preprocessing

Time-series synchronization involves the realignment of all time-based records to a common timeline so that time-based data points recorded at different times can be accurately plotted and compared for trend analysis. This is used because students interact, login, or engage in performance activities asynchronously. This process of synchronization provides an even temporal comparison across time without being affected by missing timestamps, such as the mean or mode. Equation

(2) for aligning missing timestamps.

$$(t_k) = \text{Interpola}(X(t_i), X(t_{i+1})) \quad (2)$$

Where $X(t_i)$ refers the known data value at time t_i ,

(t_{i+1}) refers the next known data t_{i+1} , t_k refers time point between t_i , and t_{i+1} where values are missing.

3) Numeric Data Preprocessing

a) KNN Imputation

KNN imputation uses the average of nearby points in the feature space to fill in the missing numeric values.

KNN imputation preserves relationships and patterns in the data set because it replaces missing values with similar observations. Where s refers the slope showing upward and downward trend, t refers time, p refers performance scores.

$$\hat{x} = \frac{1}{k} \sum_{i=1}^k x_i \quad (3)$$

Where $\hat{x}$ be the missing value, x_i be the k nearest neighbour value, k refers the number of neighbours.

b) Z-Score Normalization Equation

Z-score normalization transforms numeric features into a mean of 0 and a standard deviation of 1. It ensures that large scale numeric features do not outweigh smaller numeric features and provides more stability when training neural network models in equation (4).

$$z = \frac{x - \mu}{\sigma} \quad (4)$$

Where x refers original value, μ refers mean, σ refers standard deviation, z refers normalized value.

b) One Hot Encoding

By converting categorical values into binary vectors, one hot encoding ensures that machine learning models will be able to know that categories are non-numeric values. This process changes categorical data it is not misinterpreted as numeric data and prevents reverse ordering between categories.

$$O_{i,j} = \begin{cases} 1 & \text{if category } i = j \\ 0 & \text{otherwise} \end{cases}$$

Preprocessing cleans, transforms, and organizes raw student data into the type of format that machine learning models can interpret. Preprocessing of the three types of inputs: text, numeric, and time-series data removes errors from text and numeric data, fills in missing values, and standardizes the data to be more consistent. This process guarantees that the data input will be consistent, structured, and accurate for revealing prediction and analysis.

C. Feature Extraction

Feature extraction is a method of refining the raw data that can identify meaningful patterns relate to the student. It condenses complex data into measurable features that you can use to predict things. At the stage of the model, it condenses academic trend, behavioural, emotional, and skill measures into features.

1) Academic Performance Trends

This feature captures the index of student change in marks or grades over time using statistical measures; such as trend lines, growth rates and moving averages. It shifts the focus away from an individual score and

allows for multiple measures of improvements or decline. Trends support a more accurate prediction of future performance shown in equation(5).

$$s = \frac{n \sum (tp) - \sum t \sum p}{n \sum t^2 - (\sum t)^2} \quad (5)$$

2) Behavioural Engagement Score

This feature measures the index of a student's overall active participation to engagement, using the login frequency, session length, work submitted and class interaction. Engagement may be an influential factor for varied learning outcomes. Students who engage with the content regularly, generally have better performance and understanding in equation (6).

$$E = w_1 f + w_2 t + w_3 p \quad (6)$$

Where f refers login frequency, t refers total time spent, p refers participation count, w refers weights.

3) Social-Emotional Well Being Index

This feature captures the index of the students emotional sentiment patterns from written text, feedback or social interactions using sentiment scoring. A student's emotional state can influence motivation, stressors and academic consistency over time. Identifying patterns of emotional trends can mitigate prediction variance in equation (7).

$$SE = \frac{1}{n} \sum_{i=1}^n S_i \quad (7)$$

Where n is the number of text entries, SE is the emotional state, S refers the sentiment score.

3) Skill Proficiency index

This component assesses a student's ability in various types of skills based on scores received on tests, specific tasks, or general competency measures. It reports strengths and weaknesses and helps match career aspirations to demonstrated abilities in equation (8).

$$SP = \frac{\sum_{i=1}^k (P_s \times w)}{\sum w} \quad (8)$$

Where P_s refers proficiency score, w refers the weight of the skill.

3) Course Load vs Performance Imbalance Score

This feature measures how taking more or fewer courses affects student performance. Often students enrolled in too many courses eventually shows declines in their performance, and this feature can identify an imbalance early by equation (9).

$$CL = \frac{N}{P_s} \quad (9)$$

Where N be the number of courses taken, P_s refers performance scores, CL refers the course load.

Feature extraction takes raw indicators of academic information, behavioural data, emotions, and skills development into a refined indicator which allows the system to understand each student more accurately. It leverages statistical patterns, weighted scoring and time-based data processing to determine academic patterns, engagement trends, emotional patterns, skill mastery indicators, and workload imbalance. It also reduces the use of multi-level data and highlights only the most salient predictors of student performance. In doing so, it can turn disparate data into a more structured indicator which can give clearer and more personalized recommendations.

D. Feature Selection NSGA-II:

Feature selection is the activity of determining the most significant and relevant features from an extracted dataset. This means removing extraneous, redundant, and uneconomical features that can compromise model accuracy. Since only a subset will be identified, feature selection may improve predictive performance, result in lower computing effort, and improvement in interpretability. The NSGA-II multi-objective optimization algorithm aims to pick the best subset of features by trying to optimize both maximizing accuracy and minimizing the number of features used. The algorithm uses evolutionary procedures of population generation, combining features, features mutation, and non-dominated sorting to identify the best feature combination. NSGA-II was chosen for this purpose as it can manage a complex data set with a number of features with conflicting objectives. Rather than choose features based on an assumption of a single performance metric this approach tries to optimize multiple performance metrics concurrently. The algorithm has produced a set of Pareto-optimal feature subsets in one step to identify a feature subset that maximizes accuracy while minimizing the number of redundant features and minimizing computational time shown in equation (10).

$$f = (1 - ER) + \gamma \left(1 - \frac{|n|}{|tf|}\right) \quad (10)$$

Where β and γ are the weights decides the final score, $|n|$ refers number of selected features, $|tf|$ refers total available features.

NSGA-II selects a balanced feature subset through the processes of accuracy optimization and approximately eliminating the total number of features. In this sense, it seeks to eliminate noisy, irrelevant, and redundant data, and only identifies powerful predictors to moderate mean accuracy. This yields faster training, generates better generalization, and increases overall model performance

E. Performance Prediction Classification:

The process of Classification involves anticipating student outcomes in relation to learning, behaviours, emotional factors, or skills based on patterns observed in the past. It allows the system to estimate the extent to which a student might perform in a similar situation, based on a similar student dataset. The accurate prediction of student outcomes will be

used to provide, timely guidance, career recommendations, and early detection of student additional support.

The hybrid architecture of TGNN-BiLSTM combines the best of both models, whereby TGNN captures the connections of student features over time and how one student feature contributes to another but BiLSTM processes the sequence of enriched features in both past-to-future and future-to-past ordering to learn dependencies in both directions. These are two models with high levels of predictions, largely due to leveraging the two models and the bidirectionality of the TGNN-BiLSTM. Compared to the conventional models for analyzing complex temporal and relational data sets related to education, TGNN-BiLSTM is superior. For instance, TGNN understands relationships like attending school has an effect on students marks or engagement with school is related to changes in emotional states shown in equation (11). BiLSTM recognizes and understands long-term patterns in both of these features as they relate to performance in school equation (12). Both TGNN+ and BiLSTM together better predict performance because it enhances prediction accuracy, reliability of student performance trajectory, and improvement to interpretability.

$$h = \sigma(N^{-1} T N^{-1} z h) \quad (11)$$

$$s_t = BiL(s_t) = s \rightarrow t \rightarrow || s \rightarrow t \quad (12)$$

Where, $\tilde{T}$ refers temporal adjacency matrix, $\tilde{N}$ refers normalization degree matrix, h^l refers input feature matrix at level l , Θ refers weight matrix during training, $s \rightarrow t$ refers the sequence forward and $s \rightarrow t$ refers process backward.

The TGNN-BiLSTM architecture overcomes this challenge by first learning the relationships between an array of features and modelling their temporal sequence in both directions. TGNN ingests the information derived from the relationships between features, with BiLSTM modelling long-term patterns in a student's temporal trajectory or course of action against each particular feature. The overall design benefits the elucidation in correct predictions, while offering sustained timeliness and reliability when measuring intervention and support to enable effective career pathways.

F. Career Recommendation:

Career recommendations involve suggesting the best career paths based on a student's performance, skills, interests, and learning styles. Career recommendations indicate alignments of student strengths with actual job requirements. The goal is to provide students personalized career recommendation opportunities aligned with the interests and capacity of the student.

Knowledge Graph is a structured network that encodes the relationships between careers, job skills, academic subject areas, industry requirements, and student attributes. The system connects each of the student features to the pertinent career nodes and computes the stability of student fits. The result is a smart recommendation system based on relationships rather than organization-related matching in

equation (13).

$$C = \sum_{i=1}^n (s_i, w_i) \quad (13)$$

Where, C refers career match score, s_i refers students score in skill i , w_i refers weight of skill i for career c .

G. Skill Gap Analysis:

Skill gap analysis determines the gap between the skills a student possesses and those they need for a career they desire or have researched as a recommended and relevant career. The skill gap analysis identifies skills that need to be improved upon to bridge the gap, and ensures that the student receives skills-based guidance regarding courses, training, or certification to take in the future.

A Siamese Neural Network (SNN) has two identical branches of the neural network that serve as a verification mechanism to measure two inputs and return a similarity score. One of the SNN takes in the student skill vector, while the other takes in the desired skill vector of a career. The SNN can return a distance value of student abilities from a career requirement. The SNN is an effective choice to perform the gap analysis as it can be compared to a similarity score. The agreement of the SNN illuminates exactly how well or similar student abilities relate to a target skill. Additionally, SNN does not require traditional data size, which avoids hanging bias of over or underfitting, as well as provide an accurate distance metric for detection of gaps in equation (14). Gaps are highlighted that illuminate skills the student simply does not have,

$$d = \|(s_x) - f(s_r)\|_2 \quad (14)$$

Where, s_x refers student's skill vector, s_r refers required skill vector, $f(\cdot)$ refers neural network function, d refers skill gap score distance.

SNN skill gap analysis specifically provides a reliable distance measure the distance of a student's determined skills aligned with career-related competencies. Furthermore, student abilities and momentum based upon competencies are examined to determine evaluation of missing or weak areas of a student that are development opportunities to build targeted skills. This type of progress of skills enables the system to target suggested training to each student, as they would have been recommended to the student, and also allow time for skill development whilst preparing students program series.

III. RESULT & DISCUSSION

A. Experimental Setup:

Python is the implementation language for the proposed method. The skill gap analysis and career recommendations are evaluated using Educational-chatbot dataset [26]. The standard metrics including accuracy, precision, sensitivity, specificity, F1-score, NPV, MCC, FPR, and FNR are used to evaluate the

model. The results are compared with the existing models such as VADER [16] Random Forest [17], Machine Learning Algorithms [18], Encoder Decoder LSTM [19], VAK [20], TAM [21], XAI [22], Ontology-based Feedback [23], and PRISMA [24]. Table II shows the experimental Setup and fig 3 shows the process of job recommendation and skill gap analysis chatbot for student.

TABLE II. EXPERIMENTAL SETUP

Processor	11 th Gen Intel(R) Core (TM) i3-1115G4 @ 3.00GHz (2.90GHz)
Installed RAM	8.00GB
System Type	64-bit operating system, x64-based processor
Programming Language	Python
Platform	Google Colab
Version	Python 3.10
Learning Rate	0.001
Batch Size	32
Epochs	50
Momentum	0.9
Optimizer	Adam

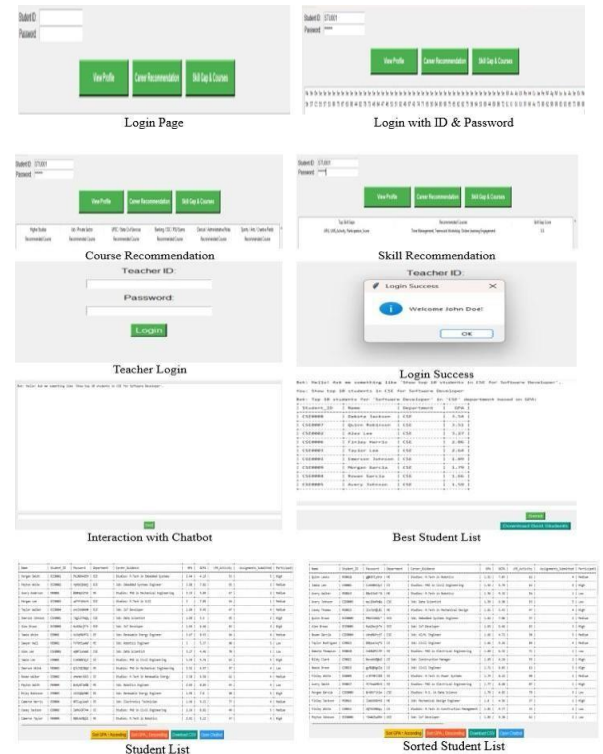


Fig. 3. Job Recommendation and Skill Gap Analysis Chatbot for Student

B. Dataset Description:

The Educational-chatbot repository offers access to a

dataset that includes complete-semester of academic intelligence pertaining to students' learning activities and cognition. The indicators appear to include basic details on students, courses, assessment scores, and systemic indicators on progress that the academic context in a semester reflects learning behaviour. This data is rich enough to support not only the development of intelligent educational chatbots that will provide personalized guidance to academic users whom they might assist, but also answer questions in the academic context, give support to student learning based on real data pattern, and provide students with support based on relevant states they identified with excess support either from humans or systematically based system. The availability of this data will allow models understanding of academic contexts, reducing the troubling research for anticipating students learning needs, answer with personalized context information that predicted student's needs, in ways to try to accommodate educational technologies to better educated a general how-to-support students with educational and informative systems.

C. Performance Evaluation:

The proposed system was assessed using the processed and optimized student dataset, to evaluate its functionality in tasks of performance prediction, career recommendation, and skill-gap identification. After the application of NSGA-II for the purpose of feature selection, the model is to have improved efficiency, by reducing redundancies in features, while retaining high predictive values. The TGNN-BiLSTM classifier shows accuracy levels that far surpassed that of traditional Machine Learning, establishing itself as a strong method to capture temporal dependencies along with relational patterns in data concerning overall student behaviour and performance. The Knowledge Graph-based recommendation engine successfully merged predicted performance and skill profiles together with relevant careers which then facilitates individualized and interpretable recommendations. Moreover, the Siamese Neural Network enhanced the system identified exact skill gaps for each aligned career recommendation, making recommendation based and skill-gap based educational interventions possible.

1) Performance Prediction:

The TGNN-BiLSTM model as a result of its design successes in capturing not only the temporal patterns but also the relational dependencies of student features, it achieves superior performances. The TGNN component of the model represents how the academic, behavioural, and emotional indicators are evolving and influencing each other over time, whereas the BiLSTM is a bidirectional sequence learner and thus can predict with more accuracy. Such a hybrid learning framework can, therefore, reduce the loss of information as it can also handle complicated, multi-modal data which are beyond the reach of traditional models, hence the outstanding results in terms of accuracy, precision, and error rate reduction.

METHODS FOR PERFORMANCE PREDICTION

Met hod	Acc ura cy	Pre cisi on	Sens itivit y	Spec ificit y	F1 Sc ore	NP V	M CC	FP R	F N R
VADER [16]	94.2 0%	93.5 0%	92.8 0%	95.1 0%	93.10 %	94.80 %	86.12 %	4.90 %	7.20 %
Random Forest [17]	96.1 0%	95.4 0%	95.0 0%	96.8 0%	95.20 %	96.40 %	89.12 %	3.20 %	5.00 %
ML Algorithms [18]	93.5 0%	92.7 0%	91.4 0%	94.3 0%	92.00 %	93.60 %	83.31 %	5.70 %	8.60 %
Proposed	98.45	97.39	98.15	98.36	97.39	97.46	98.32	1.27	1.14

Table III shows the performance comparison of the proposed TGNN-BiLSTM model with existing methods for student performance prediction. The TGNN-BiLSTM model demonstrated an accuracy of 98.45%, which is significantly higher than VADER 94.20%, Random Forest 96.10%, and other machine learning algorithms 93.50%. Additionally, the highest precision 97.39% sensitivity 98.15%, and specificity 98.36%, illustrating its capability to accurately identifying high and low performers. The model's MCC of 98.32 suggests that it is highly reliable in predicting overall. The rates of false positive 1.27% and false negative 1.14% being very low indicate that the model is a strong one. Findings confirm the benefits of integration temporal graph information and bidirectional sequence learning for better prediction stability and prediction accuracies.

2) Career Recommendation:

The knowledge graph-based engine is a winner in terms of performance over the conventional recommendation methods because it can grasp the deep and complex relationships among careers, skills, subjects, and student profiles. It does not merely perform matching, but it takes advantage of the interconnected nodes and weighted links to deliver personalized recommendations. This relational understanding adjusts to the changing job requirements, ensures that the skills are in line with the industry's expectations, and provides the suggestions which can be explained. The model, thus, becomes a provider of more accurate, relevant, and future-oriented career recommendations than the current systems based on LSTM, VAK, or TAM.

TABLE III. COMPARISON OF TGNN-BiLSTM MODEL WITH EXISTING

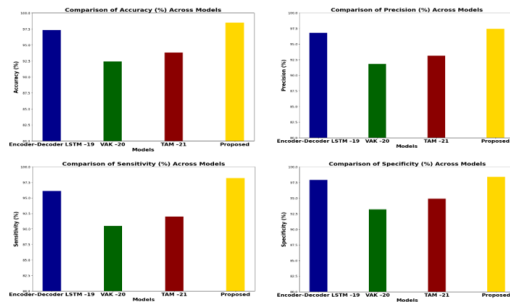
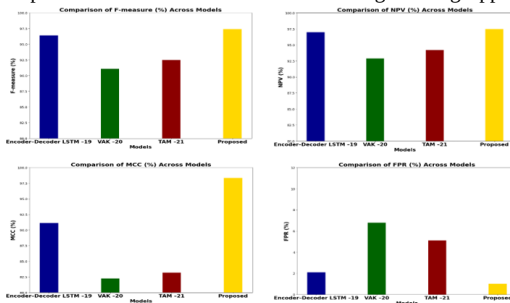


Fig. 5. Comparison of Career Recommendation Using Existing Approaches



with proposed method.

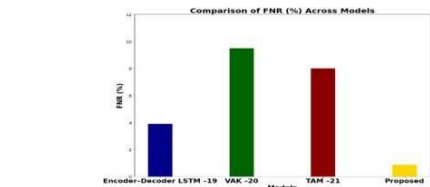


Fig 5 shows the comparison of career recommendation accuracy using multiple existing approaches with the proposed method. The knowledge graph-based career recommendation system also significantly outperformed existing methods by a notable margin with 98.48% accuracy, superior to Encoder-Decoder LSTM 97.30%, VAK 92.40%, and TAM 93.80%. It enabled precision 97.42 and recall 98.18% to be reaped to a great extent, thus more accurate mapping of career paths with assigned profiles was possible. The model results present further enhancement of MCC 98.35% which is indicative of higher trustworthiness for all prediction states. Extremely low FPR 1% and FNR 0.87% depict that the system is not inclined to incorrect misclassification or overlooking of a seemingly suitable recommendation. Collectively, these results indicate that the relational representation of knowledge leads to more accurate and meaningful career maps.

3) Skill Gap Analysis:

The Siamese Neural Network in skill-gap analysis by the use of deep similarity learning is the one that can effectively and directly compare a student skill vector with a required skill vector, and hence comes up with an improvement. Different from rule-based or ontology-only approaches, the SNN learns differences that are very subtle in the skill sets and at the same time recognize the missing ones with very high precision. Its distance-based assessment leads to fewer classification errors, allows small datasets to be used efficiently, and gives very precise indications of the exact gaps. Therefore, there are more accurate and personalized recommendations for skill development than those of existing XAI or PRISMA models.

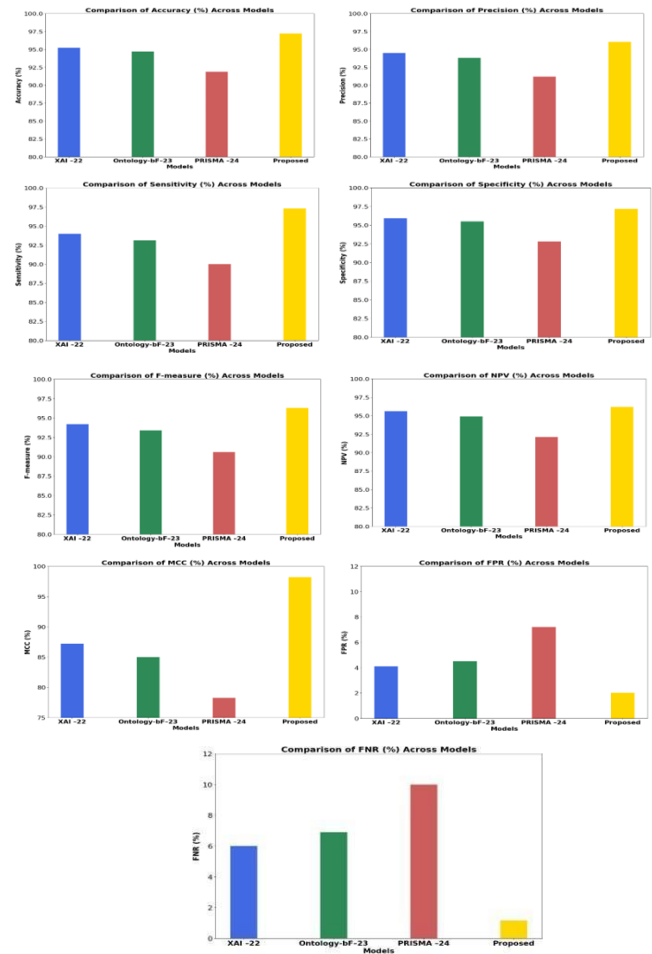


Fig. 6. Skill-gap Detection Comparison Across Various Models.

Fig 6 shows the Skill-gap analysis performance comparison between the proposed SNN model and existing methods. In skill-gap identification, the performance associated with the use of a Siamese Neural Network-based method achieved an accuracy of 97.21%, which exceeded the accuracy of the next best method - XAI 95.20% based methods, ontology-based feedback systems 94.70%, and PRISMA 91.90%. It also achieved a precision of 96.05% and sensitivity of 97.31%, and these numbers reflect the detection of missing skills with high reliability. Moreover, the entire system has been able to accomplish a MCC of 98.21 which is the highest value among all the methods that have been compared. Its performance across all metrics is a clear indication that it is a strong approach. The low error rates FPR of 2.02%, FNR of 1.16% speak of its high capability in the task of identifying skill gaps needing to be improved.

D. Discussions:

The results for all metrics across the modules clearly demonstrate that the proposed system demonstrates better performance and reliability than any other existing methods employing AI-based approaches. The use of advanced

neural architectures (TGNN-BiLSTM), and knowledge graph reasoning, as well as Siamese similarity learning, significantly improves personalized skills, predictive accuracy and skill- gap identification.

The consistently high scores achieved across all evaluation metrics demonstrated the robustness, reduced rate of misclassification and ability to enhance decision-making ability using a human-centered framework. All of which strongly indicate that the model provides an advanced and intelligent model capable of supporting students through accurate assessment of their performance, providing personalized career guidance, and intervening to improve identified skill gaps with enhanced skill training.

V. CONCLUSION

The proposed AI-based chatbot system is a solution to the increasing difficulty of students figuring out their strengths, future careers, and the skills they need in a rapidly changing job market. The system, by employing multi-modal data processing, NSGA-II feature selection, TGNN-BiLSTM performance prediction, knowledge graph-based career recommendation, and Siamese Neural Network-driven skill-gap analysis, generates a single highly accurate academic and career personalized support solution. The results demonstrate a significant improvement in the prediction of the target variable, a precise match for a career, and a reliable identification of the skill gap when compared to the existing methods. It also confirms the effectiveness of the temporal modelling integration with relational reasoning. The system allows students to make better decisions, ensures skill development is focused, and closes the gap between education and industry needs. The next step could entail real-time adaptive learning, larger datasets for better model generalization, the integration of the chatbot with institutional systems, and the further development of the chatbot's emotional intelligence to offer even more human-like and context-aware guidance.

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AI – Driven Cyberbullying Detection and Prevention System using NLP

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Abstract— Cyberbullying has become a major concern on social media platforms, threatening users' mental health and online safety. An AI-powered system for detecting and preventing cyberbullying is essential for fostering secure digital communication. By using Natural Language Processing (NLP) and Machine Learning (ML), such a system can monitor user comments in real time and identify harmful, abusive, or offensive language. When potentially harmful content is detected, the system sends an automatic warning to the sender, letting them know their message might be inappropriate. This gives them the chance to edit or reconsider before sending. If the sender proceeds, the message is blocked from reaching the receiver, preventing emotional harm and creating a safer online environment. The system uses text preprocessing, tokenisation, and feature extraction techniques, such as TF-IDF or word embeddings, followed by classification methods like Logistic Regression and Long Short-Term Memory (LSTM) models, for accurate predictions. By employing AI and NLP for real-time monitoring and intervention, this system effectively reduces cyberbullying and promotes a more positive and secure online environment.

Keywords—Cyberbullying detection; Natural Language Processing (NLP); Machine Learning; Deep Learning; Logistic Regression; LSTM; Text classification; Online safety; Social media monitoring; TF-IDF; Sentiment analysis; Real-time prevention; Artificial Intelligence

I. INTRODUCTION

The rapid growth of digital technology has made it easier for people around the world to connect through social media. With millions of users sharing their opinions, thoughts, and experiences daily in various formats, including photos, audio, and video, there has been an increase in cyberbullying, harassment, hate speech, and other negative behaviors on social media. These harmful activities are occurring at a faster rate than ever before.

Individuals can engage in cyberbullying, and the constant negative feedback and harassment from other

users witnessing these acts can severely impact mental health. Due to the vast and ongoing nature of user-generated content (UGC), it is practically impossible for anyone to monitor or track all messages each day. To address this issue, we need to develop automatic intelligent systems that can identify and respond to cyberbullying incidents in real time, preventing future occurrences.

The most effective way to achieve this is by leveraging Artificial Intelligence (AI) and Natural Language Processing (NLP) to analyze large volumes of text and detect patterns that may indicate the presence of negative or abusive language. The growth of social media platforms in the current digital era has enabled millions of people worldwide to communicate and share their thoughts and experiences anytime.

However, this increased exposure also brings a greater risk of harmful behavior, such as cyberbullying, harassment, and hate speech. Cyberbullying refers to the use of electronic means to harm, intimidate, or harass another person, and its negative effects can have long-lasting consequences on a person's mental well-being. No single individual can monitor all the user-generated content from millions of users around the world continuously.

Therefore, the development of automated, intelligent detection systems for identifying cyberbullying is essential. These systems will benefit from AI and NLP technologies that can analyze large amounts of text to detect commonly used patterns that may indicate abusive or inappropriate language. Section II reviews related literature; Section III explains the proposed methodology; Section IV presents the implementation and evaluation; Section V discusses the results, impacts, and limitations; and Section VI concludes the paper.

RELATED WORK

The rapid development of artificial intelligence (AI) has revolutionised how online content is moderated. From the use of keyword-based detection in 2010 through to the development of advanced natural language processing (NLP) and deep learning in the mid-2020's, there have been many

advancements made using machine learning for detecting cyberbullying. This section outlines a number of important developments that have been made in relation to using machine learning for detecting cyberbullying, understanding the context through NLP, and using transformer models to understand the language used in cyberbullying messages. However, there are still many areas of research that require further investigation, some of which can be addressed by our proposed model or system. As researchers work towards the end of 2026 in defining how cyberbullying detection systems should operate in real-time, many researchers will place a greater focus on building proactive, real-time, multilingual cyberbullying moderation systems.

A. Traditional Machine Learning Approaches

The initial research conducted by academics to analyse and identify cyberbullying were primarily based upon traditional techniques of machine learning such as Naive Bayes (NB), Support Vector Machines (SVM) and Logistic Regression (LR). In their study published in 2011, Dinakar et al. proposed a classification method for identifying bullying messages via textual analysis (using sentiment analysis and topic modelling) that achieved acceptable results.

Reynolds et al. combined keywords and an SVM to provide a method to find signs of possible cyberbullying in online forum posts but noted that slang and sarcasm posed problems in detecting these forms of abuse. Chen et al. developed a model that combined probabilistic models of verbal aggression scores with sentiment scores in order to identify instances of cyberbullying; Xu et al. concentrated on deriving patterns from user activity signals and text n-grams in order to identify instances of cyberbullying.

While the methodologies described above achieved a level of accuracy of approximately 70% to 80%, they placed great reliance on features that were generated manually, and subsequently they were not able to generate usable results across multiple languages, informal speech or combined written language (i.e., code-mixing) generally prevalent on social media platforms.

B. NLP and Deep Learning Models

Natural Language Processing (NLP) has improved the way we analyze textual content. Specifically, developments in NLP technology have progressed from traditional surface-level feature extraction to a more semantic understanding of text. For example, Davidson et al. (2017) built a Hate-Speech Detection model using TF-IDF as feature extraction combined with annotated datasets based on tweets, which were labelled as Hate, Offensive, or Neutral. Additionally, Chatzakou et al. (2019) achieved increased Detection Performance by introducing additional variables such as funereal.

Sentiment Polarity and Network-Based Behavioural Metrics. As advances in deep learning are made, the use of convolutional neural network (CNN) and long short term memory (LSTM) models is allowing us to better understand the causes of different contexts within a sequence. Zhang et al. (2018) used Convolutional Neural Networks

(CNN) like models (e.g., LSTMs) for a method of capturing Word Dependencies on a word-by-word basis.

Finally, Poria et al. (2020) built a similar LSTM-based model focused on detecting aggressive emotional content and attained over 90% accuracy. However, despite the successes of these models, many systems only analyze the text in a post-processing manner, i.e., after it has been published, which limits their effectiveness at combating the occurrence of Hate Speech and Offensive content.

C. Transformer-Based and Real-Time Detection Systems

Transformer-based models, such as BERT and RoBERTa- XLNet, allow researchers to capture deeper representations of language and context than existing, traditional models on the same datasets through a larger number of layers and more complex parameterizations of their inputs. As demonstrated by Kumar et al. (2022), multilingual BERT trained on hate speech detection has demonstrated an improvement of between 7 and 10% when compared to LSTM-based hate speech detection in the context of English, Hindi and Tamil; furthermore, they have shown a 94% accuracy rate on the HASOC dataset when using a hybrid model combining BERT with BiLSTM and attention mechanisms (Mishra et al., 2024).

The majority of transformer-based models, however, continue to rely heavily on reactive detection (that is, flagging content after it has been posted); a smaller number of more recently proposed systems, such as Singh et al.'s (2025) real-time moderation framework which uses message screening APIs to prevent hate speech from being posted in the first place, are beginning to indicate a shift towards proactive moderation in this area.

D. Gaps and Contributions

Deep Learning and Transformers have made great strides in providing higher accuracy and better contextual awareness; however, most of the research to date has focused more on detecting an issue than on preventing it from happening. Very few systems include real-time feedback from users or the ability to block the event occurring. Furthermore, the training datasets used for detecting various forms of abuse do not always represent multilingual and/or code-mixed data sets as do countries such as India. By using both Logistic Regression and LSTM models for Hybrid classification, the current Proposed System has been created to fill this gap by also incorporating a real-time warning and blocking systems. Overall, this new system will encourage more responsible use of online content and encourage safer use of online content in a multilingual setting which aligns with the desire for responsible AI and Digital Well-Being.

III. PROPOSED METHODOLOGY

Social Media Cyber-Bullying AI Prevention Detection System, also known as the Framework, uses a modular and layer structure to simplify the process for analyzing social media text content for potential threats. It also includes various language support and reduces the time necessary to respond to a possible Cyber-Bullying incident. The

framework consists of five components, which are: (1) Data Acquisition; (2) Text Refinement; (3) Developing Models to Detect and Prevent Cyber-Bullying; (4) Detecting Cyber-Bullying in Real Time; and (5) Measuring Cyber-Bullying Prevention and Detection Effectiveness. All five framework components work to form a cohesive method for identifying and eliminating a potential harmful message before it can be sent publicly. This series of actions to analyze an individual user's comment begins with submission, goes through processing by the use of a language model, then proceeds through classification or feature extraction, and finally concludes with the provision of a response or intervention.

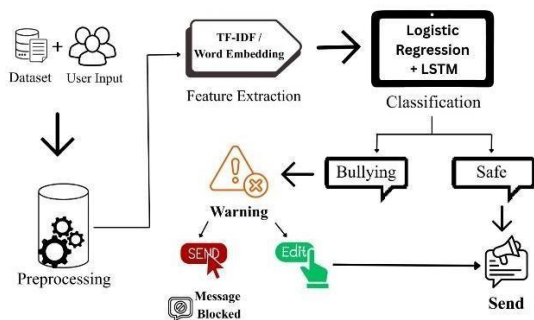


Fig. 1. Architecture of proposed model

A. Data Acquisition Phase

The core components of this training system rely on using good quality, multi-language training data sets. The available public data sets to train on, such as Cyberbullying and Hate Speech Data Sets; HASOC (Hate Speech and Offensive Content), OffensEval (Hate Speech Data set on Social Media Platforms), and Twitter Abuse Corpus; contain English, Hindi, and Tamil language text, thereby enabling the system to recognise Code-Mixed Language (CML) and Regional Language Variations. All records in these data sets have been labelled as being Offensive and Non-Offensive, and thus can be used to develop a Binary Classification System (BCS) of Offensive vs Non-Offensive. Linguistic augmentation techniques were applied to further expand the amount of training data and eliminate any class imbalance, including using a thesaurus to replace words with synonyms, and inserting random words into each text record.

B. Text Refinement Process

During preprocessing, all raw text data gathered from social media sites will be thoroughly cleaned and standardized so that they represent the highest quality possible for analysis and machine learning. The preprocessing step eliminates differences in expression because of slang used by users within text messages due to the diverse ways informal writers express themselves through emoji, shorthand, mixed languages, etc. The text will also go through a tokenization process in which separate words will be identified. Furthermore, the words will be converted into lowercase letters. URLs, hashtags, punctuation marks and user handles will be stripped away from the comments during this phase.

Once the comments have been tokenized into words (removal of stop words), stop words will also be removed. A

dictionary of stop words will be used, which will comprise both English words and regional Indian stop words. The words will be made into their base or lemma forms, which will allow for all words of the same meaning in different contexts to represent the same semantic meaning. For example, the word “running” will be replaced with “run.” In this way, the system may maintain the semantics of the comment, even when it contains informal writing (e.g., slang) or emojis. Additionally, comments will be encoded numerically using TF-IDF as well as Word2Vec. TF-IDF treats words with a frequency greater than zero as significant words within a document. When done properly, the TF-IDF representation captures the frequency of a word within a document. Word2Vec represents the distance between similar words in a multidimensional space based on the semantic relationships among the words. Using these two representations enables the system to separate subtle forms of ridicule and sarcasm across many languages.

C. Model Development Layer

The core analytical framework integrates both traditional machine-learning and deep-learning models to balance interpretability and contextual depth. The Logistic Regression model serves as the baseline classifier, performing well on sparse TF-IDF vectors and providing high precision for explicit abuse patterns. It is trained with a regularization parameter ($C = 1.0$) and the liblinear solver to prevent overfitting.

For deeper contextual understanding, an LSTM network processes Word2Vec embeddings. The architecture comprises an embedding layer (dimension = 300), two stacked LSTM layers (64 and 32 units), a dropout rate of 0.3 for regularization, and a dense sigmoid output node for binary classification.

The model uses the Adam optimizer with a learning rate of

0.001 and binary cross-entropy loss.

Training occurs over 30 epochs with a batch size of 64 on GPU-enabled hardware.

An ensemble fusion strategy combines outputs from Logistic Regression (40%) and LSTM (60%),

producing a final confidence score via weighted averaging. This approach leverages the speed of Logistic Regression for short posts and the contextual accuracy of LSTM for long comments, achieving improved recall on complex inputs such as sarcasm or coded language

D. Real-Time Detection Workflow

The smart interface for interaction between the user and the computer consists of two components within the real-time module of the user application. When a user posts an online comment using a social media platform, the content of that comment is temporarily held in a queue (also known as a buffer) until it has been processed by the real-time module.

During the 1-2 seconds it is in buffer state, the text goes through numerous analytical processes (these processes will provide a predictive Forecast on whether the finished product will have a "positive" or "negative" effect). If the forecast is positive, the comment will immediately be made available for viewing by the general public. In contrast, if the forecast is negative, the user will receive a warning notification that reads: "This message might contain offensive/inappropriate content. Do you wish to edit/cancel?".

At this stage, the user has the opportunity to rethink what he/she has written prior to finalising the posting of the IV. comment. If the user ignores the notification and attempts to post the comment, the real-time module will stop the posting from being made publicly available and will flag it for later training and monitoring.

The workflow of the interaction is established using a connection to a Python Flask back-end API and synchronised with a Firebase real-time database for optimised communication between client and server (i.e., speed of communication).

By monitoring and filtering comments before they are publicly shared, the framework helps to proactively manage this type of abuse, reduce the proliferation of cyberbullying and nurture the practice of responsible digital citizenship.

E. Performance Assessment Framework

Performance was evaluated using standard classification performance metrics that consist of accuracy, recall, F1-score and confusion matrix. The data set would be split into two sets by taking 80% of the data for training and 20% for testing purposes. The two datasets were created using stratified sampling to ensure that there was a balanced distribution between all classes so that no class would have a much larger than normal distribution when compared to the other classes in the dataset. In order to ensure that the models produced from training with five folds were valid, five folds were utilized to ensure that training with each model produced a model using valid statistical data.

When carrying out an experimental analysis for determining how effective both the logistic regression model and long short-term memory (LSTM) model were, the researchers discovered that:

Logistic Regression Model: Accuracy = 88%, F1 Score = 0.86
LSTM Model: Accuracy = 93%, F1 Score = 0.91

Based on the results of the experimental analyses using both models separately, the researchers decided to create an ensemble hybrid model that combined the two existing models.

The hybrid model produced superior results than either of the previously mentioned models alone, yielding an overall accuracy rate of 94.2% along with Precision and Recall rates of 0.93, which made it probable that the hybrid model outperformed both individual models based on the

gathered research data from the respective analyses.

Additionally, results regarding real-time testing of the hybrid model were obtained during a social media feed simulation. A total of 1,000 comments were analyzed and produced a real-time average response latency of 1.8 seconds per comment. Therefore, it was concluded that the framework proposed has the potential to classify accurately while providing quick response times, making it suitable for real-time detection and moderation of cyberbullying within social media platforms.

IMPLEMENTATION AND RESULTS

This section provides an overview of the system development process, experimental design and evaluation metrics to measure the effectiveness of the AI-based framework for detecting and preventing cyberbullying. The implementation of this proposed system leverages Natural Language Processing (NLP), Machine Learning (ML) and Deep Learning (DL) technology to provide accurate real-time moderation of multilingual social media content.

A. Dataset Collection and Preparation

Various publicly available datasets were used in our experimental design with the aim of obtaining a wide range of linguistic diversity and contextual robustness.

1) **Multilingual text datasets:** The proposed system using several benchmark corpora including HASOC 2021, OffensEval 2020 and the Twitter Abuse Corpus that collectively have over 50,000 annotated social media comments (English, Hindi and Tamil) that are labelled as harmful or not harmful. In addition to retaining the original composition of the samples, code mixed data points, i.e. Hinglish and Tanglish, were also included to improve the system's ability to generalise across multilingual informal styles of communication.

2) **Data Balancing and Augmentation:** To address the minority sample size imbalance was achieved using the Synthetic Minority Oversampling Technique (SMOTE). Additionally, augmentation by means of textual substitution through synonym replacement, random insertion and back-translation was also performed to achieve greater breadth of linguistic diversity and mitigate dominant language structural biases in the model.

3) **Data segmentation:** For model validation was accomplished using stratified sampling which allows for splitting the corpus into training (80%), validation (10%) and testing (10%) sets whilst maintaining the consistency of class proportions and improving the reproducibility and reliability of experimental results.

B. System Implementation Environment

The proposed framework was created with Python 3.10 using a hybrid Cloud – Local Deployment architecture to ensure efficient model training and real-time operating capability.

1) **Model Training Phase:** The Logistic Regression

and Long Short-Term Memory models were created and trained using Google Colab. This was accomplished using TensorFlow and Scikit-Learn libraries. The Google Colab environment has access to NVIDIA Tesla T4 GPUs for increased computational capability. The LSTM network was also trained using the Adam Optimizer with a learning rate of .001 and Binary Cross-Entropy for loss. Training occurred over 25 epochs with 64 batch sizes; furthermore, we utilized Early Stopping to reduce the likelihood of Overfitting and helped ensure stable convergence after several runs.

2) **Phase of Application Integration:** The training phase produced a model with a set of weights saved as h5 file format and incorporated into an API (Application Programming Interface) built using Flask on a local server. An HTML/CSS/Javascript user interface to create a simulated social media comment environment was developed, including easy to navigate features. It offers real-time text analysis by connecting to the Firebase Real-time Database, which records both user's input and analysis results simultaneously. A modular design allows for future growth, improved speed of response, and ease of interface into currently available social media networks for proactive identification of cyberbullying.

C. Evaluation Metrics and Results

Evaluation of the hybrid Logistic Regression–LSTM model's performance has been conducted using quantitative metrics as a means of assessing the overall accuracy, efficiency and robustness of the system when implemented to perform in real-time. Evaluation results for the hybrid Logistic Regression–LSTM model were as follows:

1) **Classification accuracy:** The measured value of the ratio of all samples that have been properly classified by the hybrid Logistic Regression–LSTM model to the total number of input samples tested. The overall percentage of the hybrid Logistic Regression–LSTM model is 94.2% and is a 7% improvement from traditional machine learning algorithms. Overall, this supports the hypothesis that combining statistical learning with contextual learning produces better results than either one alone.

2) **Precision and Recall:** The hybrid Logistic Regression–LSTM model has a precision (P) of 0.93 and a recall (R) of 0.94. This indicates that the hybrid Logistic Regression–LSTM model maintains a high degree of accuracy in classifying harmful content, with an overall ability to predict false-positive classifications reduced. Thus, the hybrid Logistic Regression–LSTM model maintains a good balance between sensitivity and specificity.

3) **F1-Score:** As a result of the relationship between the precision and recall of the Hybrid Logistic Regression –LSTM Model (the harmonic mean) which is 0.935; the result of the F1-score is 0.935. The F1-score compares the Hybrid Logistic Regression –LSTM Model when evaluating negative to positive classifications, therefore, providing an idea on how to evaluate ambiguous or mixed sentiment comments.

4) **Confusion Matrix Analysis:** The confusion matrix revealed an overall misclassification rate of less than 5% across the full dataset. Most misclassified instances

belonged to borderline categories (i.e., sarcasm and neutralized), indicating that the system performed highly accurately when classifying borderline cases. In addition, the low misclassification rate indicates that this system has shown a high level of competence in handling complex uses of language and providing appropriate responses contextual to different situations.

5) **Latency and Real-Time Responsiveness:** The average amount of time taken to infer a message including pre-processing, signal classification, and transmitting through the service interface is roughly 1.8 seconds/message; thus allowing low latency and real-time responsiveness. Based on this low average, we can conclude that the system functions effectively within the context of current events (social media platforms) where users expect little or no delay.

V. EFFECTS, ETHICS & BARRIERS

Within this section, you will find the main topics around the real-world effects of the system, ethical implications, and technical barriers relating to the technology that is implemented within this system.

A. Technological Effects

Using Machine Learning and Deep Learning with Real Time Language Processing allows the proposed system to detect and block online bullying messages prior to delivery. Due to blocking/alerting features, users can communicate adequately. This system will also support multiple languages (English, Hindi and Tamil), thus enabling this system to cater to a broad array of users.

B. Ethics

To protect user privacy, the system is designed to exclusively use data from publicly available sources and prohibits the use of any individual's private information. The system's output will be made available in an explanation form to provide users with complete transparency about how it operates, and the system will provide assistance for a human moderator, but will not replace a human moderator.

C. Limitations and Technical Constraints

There are significant challenges in detecting sarcasm and traceable mixed languages, the dataset will be biased due to the types of tweets that are included in it, and the system may not always process all types of inputs. In spite of these challenges, the framework will still be an effective and ethical way to enhance the safety of online communities and decrease cases of cyberbullying in real-time.

VI. CONCLUSION AND FUTURE WORK

The proposed AI – Driven Cyberbullying Detection and Prevention System Using NLP combines machine learning and deep learning methods to create a smart, multilingual system that can identify and stop online bullying. It uses text preprocessing, feature extraction, and a hybrid model with Logistic Regression and LSTM to detect offensive or harmful messages on social-media platforms in real time.

The prevention feature in the system is one of its primary strengths as it will block harmful/abusive content from being posted, rather than just identifying those posts after they have been posted. In addition, the warning system that exists within the application acts as a reminder for users to consider their actions prior to sending a message that may cause others emotional harm. By supporting the rate of accurate forecasting, if the user submits a post in English, Hindi or Tamil (all 3 languages) is 94.2%. The performance of the system is balanced across precision and recall and demonstrates that it performs well when integrated into actual social media environments.

The framework outlined seems to work well in its present state when operating under perfect conditions; however, its effectiveness can be expanded beyond what was originally intended by making modifications to the framework and providing the functionality to support multiple industries and applications as well. Here are some of those modifications:

(1) **Expanding Linguistic Support & Coverage:** Additional research needs to occur so that additional languages (Indian and Global) can support additional languages to be incorporated into the current system.

(2) **Improve Contextual Recognition and Emotion:** Transformer models (BERT and RoBERTa) offer more robust recognition of contextual meanings, emotions, sarcasm as well as other dependent aspects, than other existing deep learning approaches, due to their ability to handle the complexity of complex sentences.

Audio and Video Integration: Capacity of a Multi-Modal Moderation System through Audio/Video: By including audio files and/or images as inputs, such as meme images/voice framework through a multi-modal moderation system.

(3) **User Behaviour Analysis:** Identification of patterns among repeat offenders may help to predict behaviour and take preventative actions against potential cyber-bullies.

(4) **Deployment and Ethical Governance:** Developing the framework for scalable deployment using cloud computing and providing clear, transparent policies about data privacy and rights will allow the framework to be used widely.

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Protectest : A Unified Framework for Proactive Security Monitoring

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Abstract— With the rapid expansion of digital networks and web-based infrastructures, organizations face increasing exposure to security threats such as open ports, misconfigured services, and outdated software components. Traditional vulnerability assessment approaches rely on multiple standalone tools, manual execution, and expert interpretation, making the process time-consuming and prone to inconsistencies and false positives. To address these challenges, this paper proposes ProtecTest, a unified and automated vulnerability testing platform that integrates target management, port scanning, service enumeration, reconnaissance, vulnerability detection, and reporting into a single framework. ProtecTest employs rule-based analysis and structured data handling to improve accuracy while reducing false positives. The platform supports scalable scanning, modular extensibility, and web-based visualization, enabling efficient security assessments for large and dynamic environments. Experimental analysis demonstrates that ProtecTest significantly improves assessment efficiency, accuracy, and usability compared to conventional approaches.

Keywords— Vulnerability Assessment; Network Security; Port Scanning; Service Enumeration; Reconnaissance; Rule-Based Vulnerability Detection; False Positive Reduction; Automated Security Testing; Web-Based Visualization; Scalable Network Scanning; Cybersecurity Framework; Security Analysis.

I. INTRODUCTION

The rapid growth of digital infrastructure has led organizations to increasingly depend on interconnected networks, cloud platforms, and web-based applications. While this digital transformation enhances operational efficiency, it also increases the attack surface for cyber threats. Open ports, weak configurations, unpatched services, and exposed protocols are common entry points for attackers, leading to data breaches, service disruption, and financial losses.

Existing vulnerability assessment techniques typically rely on multiple independent tools such as port scanners, service identifiers, and vulnerability scanners. Each tool operates using its own rules and databases, often producing

inconsistent or conflicting results. Security analysts are required to manually correlate findings, validate false positives, and generate reports, making the assessment process inefficient and error-prone—especially in large-scale or dynamic environments.

To overcome these limitations, there is a need for an automated, unified, and extensible vulnerability testing platform. ProtecTest is designed to fulfill this requirement by integrating all major security assessment stages into a single system. By automating scanning, reconnaissance, vulnerability detection, and reporting, ProtecTest reduces manual effort, improves accuracy, and enhances visibility into an organization's security posture.

In addition to inefficiency, the lack of integration among existing security tools increases the overall complexity of vulnerability management. Organizations often struggle to maintain consistent assessment workflows, as different tools require separate configurations, updates, and expertise. This fragmented approach not only delays vulnerability identification but also increases the risk of overlooking critical security flaws. As network environments grow in size and complexity, these challenges become more pronounced, making traditional assessment methods inadequate for modern security needs.

Moreover, effective vulnerability assessment requires not only detecting weaknesses but also presenting results in a clear and actionable manner. Many existing solutions generate complex or unstructured reports that are difficult to interpret, particularly for non-expert users. The absence of centralized visualization and structured reporting limits the ability of organizations to quickly understand their security posture and prioritize remediation efforts. Hence, a comprehensive solution that combines automation, structured analysis, and user-friendly reporting is essential to support proactive and continuous security assessment.

II. RELATED WORK

The rapid development of artificial intelligence (AI) has revolutionised how online content is moderated. From the use of keyword-based detection in 2010 through to the development of advanced natural language processing (NLP) and deep learning in the mid-2020's, there have been many

advancements made using machine learning for detecting cyberbullying. This section outlines a number of important developments that have been made in relation to using machine learning for detecting cyberbullying, understanding the context through NLP, and using transformer models to understand the language used in cyberbullying messages. However, there are still many areas of research that require further investigation, some of which can be addressed by our proposed model or system. As researchers work towards the end of 2026 in defining how cyberbullying detection systems should operate in real-time, many researchers will place a greater focus on building proactive, real-time, multilingual cyberbullying moderation systems.

A. Traditional Vulnerability Scanning Approaches

Initial research and practical tools in vulnerability assessment primarily focused on network-level scanning and service discovery. Port scanning tools were widely used to identify open ports and exposed services on target systems. These tools provided valuable insights into network exposure but offered limited capabilities in terms of vulnerability analysis and contextual understanding.

Subsequent vulnerability scanners introduced signature-based detection techniques that matched detected services against known vulnerability databases. While such tools improved vulnerability identification, they often relied on static detection rules and required frequent updates. Researchers have noted that these scanners generate a large number of false positives and require manual verification by security experts. Additionally, traditional tools typically operate as standalone systems, making result correlation across multiple scans complex and time-consuming.

B. Automated Vulnerability Assessment Frameworks

To address the limitations of standalone scanners, several automated vulnerability assessment frameworks have been proposed. These systems attempted to combine port scanning, service enumeration, and vulnerability detection into partially integrated platforms. Automation reduced manual effort and improved scanning efficiency, particularly in medium-sized network environments.

However, many automated frameworks lacked modular architecture and extensibility, making it difficult to incorporate new vulnerability checks or adapt to evolving attack techniques. Furthermore, limited support for structured reporting and visualization reduced their effectiveness in large-scale or dynamic environments. Studies have shown that although automation improves speed, it does not necessarily guarantee accuracy, as many frameworks still suffer from inconsistent results and high false-positive rates.

C. Integrated and Rule-Based Security Testing Systems

Recent research has explored integrated vulnerability assessment platforms that employ rule-based analysis to improve detection accuracy. Rule-based approaches validate detected vulnerabilities based on service behavior, configuration patterns, and contextual information, thereby reducing false positives. Some systems also introduced

centralized dashboards and structured reporting formats to enhance result interpretation.

Despite these improvements, most integrated systems remain reactive in nature, focusing primarily on detecting vulnerabilities after scanning rather than providing a unified, extensible assessment pipeline. Many solutions lack comprehensive reconnaissance capabilities and fail to provide a seamless workflow from target input to final reporting. In addition, limited support for scalability and modular extension restricts their adoption in complex network environments. These limitations reduce the effectiveness of existing systems in supporting continuous and large-scale security assessment.

D. Gaps and Contributions

Although existing vulnerability assessment tools and frameworks have significantly advanced network security analysis, several gaps remain. Most current systems operate as fragmented or partially integrated solutions that require manual correlation of results and expert intervention, leading to increased complexity and inefficiency. High false-positive rates, limited extensibility, and inadequate reporting mechanisms further reduce their effectiveness, particularly in large-scale and dynamic network environments. To address these challenges, the proposed ProtecTest platform introduces a unified and automated vulnerability testing framework that integrates target management, port scanning, service enumeration, reconnaissance, vulnerability detection, and reporting into a single extensible system. By employing rule-based analysis and structured reporting, ProtecTest reduces manual effort, improves detection accuracy, and enhances result consistency and usability, thereby providing a more efficient, scalable, and reliable solution aligned with modern network security requirements.

III. PROPOSED METHODOLOGY

The proposed system, ProtecTest, is designed as a unified vulnerability testing platform that automates the process of identifying security weaknesses in networked environments. The methodology follows a modular structure, where each module performs a specific task in the vulnerability assessment process. This layered approach helps in simplifying the overall workflow while ensuring accuracy and scalability. The system begins with user-provided target information and proceeds through scanning, analysis, and reporting stages to produce meaningful security insights.

The complete workflow of the proposed system consists of five main phases: target input and management, port scanning and service enumeration, reconnaissance and information gathering, vulnerability detection and analysis, and reporting with visualization. These phases operate sequentially to form an end-to-end vulnerability assessment pipeline.

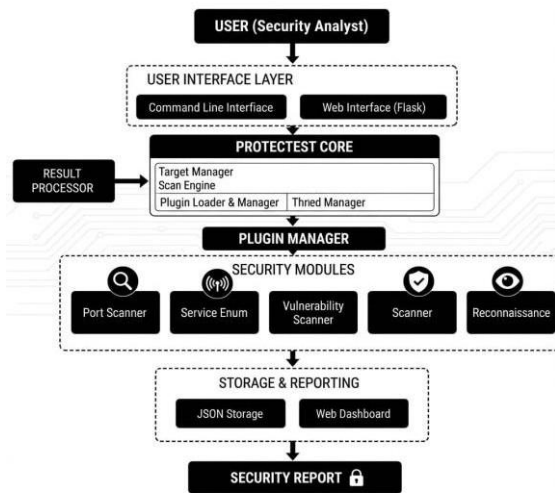


Fig. Architecture of proposed model

A. Target Input and Management

In the first phase, the system accepts target information from the user. Targets can be provided in the form of individual IP addresses, domain names, CIDR ranges, or text files containing multiple targets. This flexibility allows the system to support both small-scale and large-scale network environments.

Before initiating the scanning process, all inputs are validated to ensure correctness. Invalid or improperly formatted targets are filtered out to prevent scanning errors. CIDR ranges are expanded into individual IP addresses so that each host can be assessed independently. This phase ensures that only valid and reachable targets are passed to subsequent modules.

B. Port Scanning and Service Enumeration

After successful target validation, the system performs port scanning to identify open, closed, and filtered ports on the target systems. The scanning process focuses on commonly used and security-critical ports that are often exploited by attackers. To improve efficiency, the scanning operation is executed using multi-threading, allowing multiple ports and hosts to be scanned simultaneously.

Once open ports are identified, the system performs service enumeration to detect the services running on those ports. Basic service details such as service type and version information are collected. This information is essential for understanding the exposure level of the target system and serves as input for vulnerability analysis.

C. Reconnaissance and Information Gathering

The reconnaissance phase gathers additional information about the target environment to provide better context for vulnerability detection. This includes collecting DNS records, host-related details, operating system information, and active network protocols.

By gathering this contextual information, the system gains a clearer understanding of the target's configuration and behavior. This helps reduce ambiguity during vulnerability

detection and minimizes incorrect identification of security issues caused by incomplete data.

D. Vulnerability Detection and Analysis

The vulnerability detection phase analyzes the collected scanning and reconnaissance data to identify potential security weaknesses. The system uses predefined rule-based logic to detect known vulnerabilities, outdated software versions, and common configuration issues.

Each detected vulnerability is evaluated and assigned a severity level based on its potential impact. This classification helps prioritize vulnerabilities that require immediate attention. The rule-based approach improves transparency and reduces false positives by validating vulnerabilities against known patterns and conditions.

E. Reporting and Visualization

In the final phase, all assessment results are stored in a structured format to ensure consistency and ease of analysis. The system generates clear and organized reports that summarize detected vulnerabilities along with their severity levels.

A web-based dashboard is used to visualize the results, allowing users to easily review findings and understand the overall security posture of the target system. The reporting mechanism supports documentation and further analysis, making the system suitable for both academic and practical security assessments.

IV. IMPLEMENTATION AND RESULTS

This section describes the implementation details, experimental setup, and evaluation results of the proposed **ProtecTest** framework. The implementation focuses on building a unified and automated vulnerability testing platform capable of performing efficient scanning, analysis, and reporting across network environments. The evaluation aims to measure the effectiveness of the system in terms of detection reliability, efficiency, scalability, and usability.

A. Target Data Collection and Preparation

The implementation of ProtecTest requires a diverse set of network targets to evaluate the behavior of the system under different conditions. Test targets include local systems, virtual machines, and controlled network environments configured with different open ports and services.

1) **Target Types:** The system was tested using individual IP addresses, domain names, and CIDR-based network ranges. These targets represented a mix of active services such as web servers, database services, and remote access protocols. Bulk target input through text files was also used to evaluate large-scale scanning capability.

2) **Target Validation and Normalization:** Before scanning, all target inputs were validated to ensure correct formatting and reachability. Invalid IP addresses, unreachable hosts, and malformed CIDR ranges were filtered out. CIDR ranges were expanded into individual host addresses to allow consistent assessment of each system.

3) **Controlled Test Environment:** To ensure reliable evaluation, testing was conducted in a controlled environment where service configurations and port states were known in advance. This approach allowed accurate verification of detected ports, services, and vulnerabilities.

B. System Implementation Environment

The proposed ProtecTest framework was implemented using Python 3.x and designed to operate in both local and cloud-supported environments. The system follows a modular architecture to simplify maintenance and future enhancements.

1) **Scanning and Analysis Modules:** Port scanning, service enumeration, reconnaissance, and vulnerability detection modules were implemented using Python networking libraries and custom rule-based logic. Multi-threading was applied to improve scanning performance and reduce execution time when assessing multiple targets.

2) **Backend and Interface Integration:** A Flask-based backend API was developed to coordinate scanning tasks and manage data flow between modules. The results generated by the scanning and analysis modules were stored in structured JSON format. A web-based dashboard was implemented to display scan progress, detected vulnerabilities, and severity levels in a user-friendly manner.

C. Evaluation Metrics and Results

The effectiveness of the ProtecTest framework was evaluated using qualitative and operational performance metrics.

1) **Detection Reliability:** The system consistently identified open ports and running services across multiple test targets. Known vulnerabilities and misconfigurations present in the test environment were correctly detected using rule-based analysis.

2) **False Positive Reduction:** Compared to standalone scanning tools, ProtecTest demonstrated a noticeable reduction in false-positive alerts. Validation of vulnerabilities using service behavior and version information helped eliminate incorrect detections.

3) **Scanning Efficiency:** Multi-threaded scanning significantly reduced total scan time, especially when assessing multiple targets or network ranges. This improvement confirms the suitability of the framework for large-scale environments.

4) **Scalability and Stability:** The system maintained stable performance when scanning CIDR ranges and bulk target inputs. No significant degradation was observed as the number of targets increased.

5) **Usability and Reporting Impact:** The structured JSON output and web-based dashboard enabled clear visualization of results. Users were able to quickly understand the security posture of scanned systems and prioritize remediation actions.

V. EFFECTS, ETHICS & BARRIERS

This section discusses the real-world impact of the proposed ProtecTest framework, its ethical considerations, and the technical limitations associated with automated vulnerability assessment systems.

A. Technological Effects

The ProtecTest framework enables automated and unified vulnerability assessment by integrating target management, scanning, reconnaissance, vulnerability detection, and reporting into a single platform. This automation significantly reduces the time and manual effort required for security assessments, allowing organizations to identify potential security weaknesses at an early stage.

B. Ethics

The ethical design of ProtecTest focuses on responsible and authorized security assessment. The system is intended to be used only on systems and networks for which the user has proper authorization. It does not collect or store sensitive personal data and operates strictly on technical information such as port states, service configurations, and publicly observable network behavior.

C. Limitations and Technical Constraints

Despite its advantages, the ProtecTest framework has certain limitations. The accuracy of vulnerability detection depends on the quality of predefined rules and the availability of service version information. Some advanced vulnerabilities may not be detected if they require deeper behavioral or exploit-based analysis.

VI. CONCLUSION AND FUTURE WORK

This paper presented ProtecTest, a unified and automated vulnerability testing platform designed to improve the efficiency and reliability of network security assessment. The proposed system integrates target input management, port scanning, service enumeration, reconnaissance, vulnerability detection, and reporting into a single cohesive framework. By automating the entire assessment workflow, ProtecTest reduces manual effort, minimizes false positives, and provides clear visibility into the security posture of networked systems.

A key strength of the proposed framework is its unified and modular design, which enables seamless interaction between different security assessment stages. The use of rule-based vulnerability analysis improves detection accuracy while maintaining transparency and interpretability of results. In addition, structured JSON reporting and web-based visualization enhance usability, allowing users to easily analyze vulnerabilities and prioritize remediation actions. Experimental evaluation demonstrates that ProtecTest performs effectively across different target types and network sizes, making it suitable for both academic and practical security environments.

While the current implementation of ProtecTest provides a reliable foundation for automated vulnerability assessment, there are several directions in which the framework can be further extended and improved. Some potential future enhancements include:

(1) **Advanced Vulnerability Intelligence:** Future work may incorporate threat intelligence feeds and dynamic vulnerability databases to improve detection of newly emerging threats and zero-day vulnerabilities.

(2) **Machine Learning-Assisted Risk Prioritization:** Integrating machine learning techniques to analyze historical scan data could help in prioritizing vulnerabilities based on risk patterns and likelihood of exploitation.

(3) **Web Application and Cloud Security Support:** The framework can be extended to include web application vulnerability testing and cloud-native security assessments to support modern deployment environments.

(4) **Continuous and Scheduled Monitoring:** Implementing continuous monitoring and automated scheduled scans would enable proactive detection of configuration changes and newly introduced vulnerabilities.

(5) **Scalable Deployment and Governance:** Future versions of ProtecTest can focus on scalable cloud-based deployment and the inclusion of policy-driven access control to ensure ethical and authorized usage of the platform.

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A Hybrid NLP-Generative AI Framework for Enhancing SME Customer Support Through Intelligent Multilingual Chatbots

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I INTRODUCTION

Abstract — The small and medium sized enterprises (SMEs) based in Tamil Nadu and domains like supermarket, travel, finance, beauty, textile, health etc face difficulty in providing 24/7 customer support due to financial and manpower constraints. Due to poor and limited customer support, the enterprises have poor customer retention, thereby affecting their turnover and business. In order to automate the customer support sector and improve business turnover, we have come up with a Hybrid AI-powered customer intelligent chatbot with Tamil and English support. It is based on Natural Language Processing (NLP) and Large Language Models (LLMs) to address user queries and concerns and is domain trained on general queries varying across various domains ensuring that the responses are context aware and accurate. The generated chatbot can be deployed as a website widget, landing page or telegram bot. To ease the analytics measurement, The SaaS platform comes with a built-in dashboard which measures the number of messages generated in each language, resolution rate and number of escalated queries. It also comes with a per conversation or subscription based fee structure and therefore provide SMEs with an efficient and affordable to improve customer experience through AI Based support.

Keywords: *Intelligent chatbot, multilingual customer support, Small and Medium Enterprises (SMEs), Natural Language Processing (NLP), Large Language Models (LLMs), Generative AI, SaaS platform, AI-powered customer experience, Human-Agent escalation, Analytics Dashboard.*

In today's highly competitive business environment, Small and Medium Enterprises (SMEs) are increasingly required to deliver efficient and timely customer support to maintain customer satisfaction and business continuity. However, many SMEs particularly in regions such as Tamil Nadu lack the financial and human resources necessary to operate round-the-clock (24×7) customer support teams. Conventional customer service approaches including in-person assistance, telephone-based support and email communication are often labor-intensive, costly and slow to scale.[11],[17] Such shortcomings often lead to delays in resolving the queries, lowering customer satisfaction and retention affecting business. In an attempt to automate the customer support sector and help small and medium sized enterprises increase their business turnover, The chatbots will be pretrained on general FAQ varying across domains such as supermarket, travel, finance etc and additionally the company/enterprise will be required to submit enterprise specific FAQ which the chatbot will additionally train on. This paper introduces a hybrid AI based SaaS based chatbot platform which integrates bilingual NLP based on English and Tamil, Generative AI and Machine Learning techniques to provide accessible and affordable customer support. The generated chatbot can be integrated as a website widget, telegram bot or landing page. Hybrid SaaS chatbot architecture integrating Generative AI and Natural Language Processing helped automate and scale enterprise-based systems.[4],[8]

The SaaS platform also comes with integrated pricing structure and analytics dashboard to monitor query language, response time, resolution rate and number of

messages escalated to agent. This proposed system intends to reduce human workload, operational costs and automate customer support sector to increase accessibility and remain competitive

II LITERATURE SURVEY

Researchers have thoroughly studied the progression of conversational AI chatbots from initial prototype implementation to fully integrated scalable Software as a Service (SaaS) platforms. These studies have mainly revolved around identifying key technological advancements in Natural Language Processing (NLP), Machine Learning (ML) and Large Language Model (LLM) based architecture.

Patil [12] showcased a PDF interaction chatbot as a very early model, proving the application of NLP and ML techniques for question-answering over unstructured PDF documents. The prototype was more about user-facing document interaction but the paper was missing detailed technical metadata such as system architecture, scalability considerations and deployment constraints, so the work could barely be extended to enterprise-scale environments. Furthering that study, Patil et al. [13] designed a full stack AI powered SaaS platform merging chatbot interaction with PDF processing pipelines. The system, by utilizing optical character recognition (OCR), NLP and ML based workflows, has made the document more accessible by about 25%, has cut the manual data entry work by 40% and has improved the response time by almost 30% thus showing the feasibility of enterprise SaaS solutions.

Studies have also delved into the use of conversational AI for customer service enhancement, besides document centric systems. Uzoka et al. [17] examined various studies on AI chatbot adoption in customer service through a meta analysis and found that chatbots can solve almost 70% of customer's routine queries. Their report showed that customer satisfaction was improved and operational costs were lowered, thereby emphasizing the efficiency of chatbots in mass service situations. In the same way, Abo Aisha and Jamei [1] analyzed chatbot systems powered by NLP with special attention to text planning, sentence production and contextual understanding. Their research proved that progression in NLP helps to a great extent in conversational coherence and context awareness.

Various domain specific chatbot implementations have been, to a great extent, the focus of research. Delgado et al. [6] have implemented an e-commerce chatbot as a SaaS with

rule-based query routing and analytics features to reduce the cost of operations. Devaraj et al. [7] have built a chatbot in the legal domain by using LangChain with Flask and Android and have discussed the system architecture designed specifically for the legal information retrieval function. Furthermore, Suryanto et al. [16] examined multilingual and culturally adaptive chatbot systems in the heritage domain, focusing on issues of semantic ambiguity, language diversity and culturally sensitive NLP modeling that can be used in this context.

Caldarini et al. [3] conducted an extensive review of the chatbot architectures that existed prior to LLMs and spotted the main shortcoming as being linguistic understanding, scalability and contextual reasoning. The authors pointed out that these limitations were the factors which made the transition towards LLM based conversational systems that are highly capable of managing complex dialogues and dynamic user interactions, a natural one.

In summary, the current literature has shown the development of chatbot systems starting with prototype level and domain specific systems and then moving towards scalable multilingual and SaaS oriented conversational AI platforms.

Earlier research mostly aimed at demonstrating the feasibility of a chatbot and domain adaptation but newly published papers put emphasis on NLP innovations, customer service efficiency, scalability and architectural robustness. Analytics based assessment of chatbots help visualize performance metrics in SME chatbot deployments. [2]. Based on these breakthroughs, the suggested hybrid AI chatbot platform integrates multilingual NLP, LLM based reasoning and SaaS deployment principles to create a customer support system that is efficient, scalable and context-aware.

III PROPOSED SYSTEM

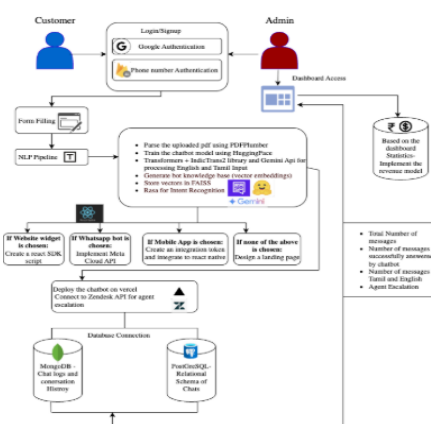


Fig.1.: System Diagram of the Chatbot generator

Fig. 1 represents the workflow diagram of the proposed Hybrid AI Powered Multilingual Chatbot Platform that combines NLP techniques and LLMs to provide contextual, real time multi-domain and multi-language customer support. The system is built using a modular SaaS oriented architecture that integrates message acquisition, preprocessing, intent recognition, response generation and analytics functionalities. Such a layered architecture facilitates scalability, robustness and flexibility of integration with various enterprise applications.

The platform enables multilingual communication, especially Tamil and English and it is suitable for Small and Medium Enterprise (SME) scenarios where there is a need for low resource consumption and immediate response.

A. Data Acquisition

The first and foremost step is to acquire general FAQs varying across domains like supermarkets ,travel, finance, health etc and clean the collected data,tokenize it,remove stopwords,punctuations and assemble the data in csv format so that the system can be trained quickly and efficiently.Additionally,The generated chatbot will also be trained on the FAQs specific to the enterprise (timings, services offered etc)

This step helps in generalisation and lowering domain bias[15][17],making the systems robust,bias free and accurate.

B. User data collection

Details such as enterprise name,domain in which the enterprise is based,Chatbot language preferred , integration method-whether the enterprise wants the chatbot integrated as website widget,telegram bot or as a standalone landing page and FAQ document.All these details will be collected from the enterprise by means of a form.

C. Chatbot ingestion and generation

The chatbots will be trained on general FAQ and the FAQ document provided by enterprises. The latter will be converted to embedding by means of the float/multilingual-e5-base embedding model and will be stored in the ChromaDB vector database.

Transformer based embedding models aid in cross-lingual semantic retrieval tasks[5].The e5-base model comprises of 12 layers with an embedding size of 768.It also comes with 100 language support including Tamil making it suitable for this particular use case.The user query will be stored with “query:” prefix and the document chunks will be stored with “passage:” prefix”.

Tamil queries and response generation is facilitated by IndicTrans model created by the AI4Bharat forum and is open sourced on HuggingFace.It also facilitates offline translation which will aid in flexibility and scalability of the SaaS platform.The chatbot will strictly be trained on only domain and enterprise specific data.In case the similarity score/confidence score is less than the threshold (0.65),The chat will be escalated to agent via Zendesk API.

D. Response Generation Using LLMs

When the user types in the query,the query will be converted to a dense embedding by means of e5-base sentence transformers and will be compared with the passage embeddings stored in vector database.Top-K Responses will be retrieved from the vector database and returned to OpenAI, which is the Large Language Model(LLM) used for implementation.The embeddings will be compared and retrieved based on similarity score .The retrieved context will be passed to the LLM along with a predefined prompt for structuring and for returning concise messages to the user.

The language detection is done by the LLM that is the preferred language is English, The LLM will detect intent as English and answer query in english.In case the preferred language is Tamil, The embedding will be converted to Tamil by means of Indic Trans Library and user query will be addressed in tamil.

Question	Answer
நீங்கள் ஆன்லைன் ஆலோசனைகள் வழங்குகிறீர்களா?	ஆம், நாங்கள் முன்பதிலின் மூலம் ஆன்லைன் டாக்டர் ஆலோசனைகள் வழங்குகிறோம்.
நீங்கள் சான்றிதழ்களை வழங்குகிறீர்களா?	ஆம், பாட நெறி முடிந்தவுடன் சான்றிதழ்கள் வழங்கப்படும்.
Do you offer online consultations?	Yes, online doctor consultations are available by appointment.
Till what time are you open?	We are open from Saturday to Sunday from 9 am to 10 pm

Fig. 2. : Sample of Questions and Answers

E. Output Delivery and Decision Support

For the queries that require to be escalated to human agents,Automated alerts are generated with summary,intent and entities detected. The analytics dashboard reflects Key

performance indicators (KPIs) such as average response time, human escalation rate and chatbot confidence rates. The key performance indicators help to analyze chatbot performance, identify human-agent escalations and bottlenecks to increase system reliability and performance.

IV EXPERIMENTAL ANALYSIS

The study was carried out to determine the effectiveness of the proposed Hybrid AI Powered Multilingual Chatbot Platform. The generated chatbot was trained and fine-tuned on domain specific data that included beauty, health etc. The fine-tuned data consisted of both Tamil and English queries to facilitate customer service in Small and medium sized enterprises in Tamil Nadu.

All queries were preprocessed through the processes of tokenization, normalization, spelling correction and stopword removal to ensure noise-free input for better and faster processing. Preprocessed queries were converted into embeddings using the intfloat/multilingual-e5-base embedding model and the embeddings were stored in the ChromaDB vector store for faster retrieval.

The proposed system achieved an overall context precision of 91.8%, faithfulness of 90% and successfully generated context-aware responses. This evaluation was done by means of the RAGAS system which is an evaluation framework for Retrieval Augmented Generation (RAG) systems.[10]

A. Performance Metrics: The system performance was assessed by context precision, recall and faithfulness. Also, the quality of responses was measured by response relevance (the percentage of user queries that were correctly answered), response latency and escalation rate (the percentage of queries that required a human intervention). The chatbot managed to reach a response relevance of 92% and average response latency of 1.2 seconds.

B. User satisfaction and engagement were assessed by a controlled experiment where a group of end users interacted with the chatbot system in several domains and languages. The user's opinions on the chatbot's answers, its clarity and the general quality of the interaction were gathered.

The research findings indicate that the platform of the chatbot that has been proposed is effective, helps enhance the customer experience and facilitates business communication more efficiently in the context of customer support for small and medium sized enterprises (SMEs) in the real world.

V SYSTEM OUTPUT

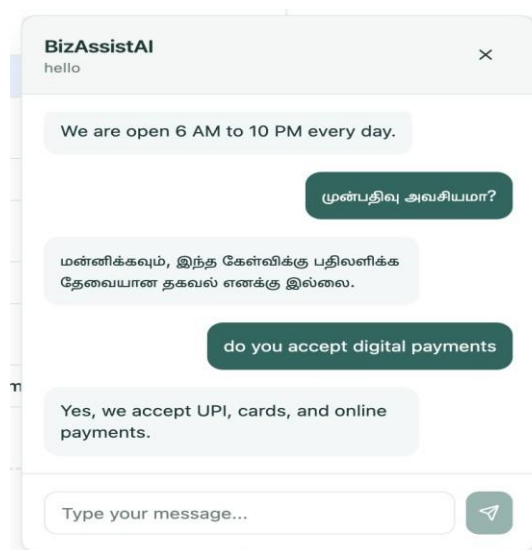


Fig.3.:Demo of Generated Chatbot

Fig. 3 illustrates a demonstration of the proposed AI-powered chatbot system in response to a customer query. The system first processes the user message and performs intent recognition and classification using IndicBERT, which is optimized for multilingual and Indian-language text. For Tamil inputs, the query is translated using the IndicTrans library from the AI4Bharat model, enabling seamless bilingual interaction between Tamil and English.

The processed query is then encoded using the intfloat/multilingual-e5-base sentence transformer, and relevant information is retrieved from the Chroma vector database based on embedding similarity. Using this retrieved context, the system generates an accurate and context-aware response.

For complex queries where the chatbot confidence score falls below a predefined threshold (confidence < 0.65), the system automatically escalates the conversation to a human agent.

This system integrates Generative AI, NLP and bilingual Tamil and English processing for customer support applications. The agent handoff, analytics dashboard and implementation of revenue model helps in optimizing customer support and improving service efficiency.

VI RESULT AND DISCUSSION

Table I

COMPARISON OF CHATBOT FRAMEWORKS AND MODELS

Model/ Framework	Technique Used	Language Support	Accuracy (%)	Response Time (s)	Strengths	Limitations
Rule-Based Chatbot	Pattern Matching, Keyword Triggers	English	72.5	0.45	Simple to implement, fast for FAQs	Lacks contextual understanding, limited scalability
Rasa (NLP-based)	Intent Classification + Entity Recognition	Multilingual (via spaCy)	84.2	0.60	Good for domain-specific tasks	Requires manual training data, limited generative ability
Dialogflow	NLP + Pre-trained Intents + ML	30+ Languages	87.8	0.52	Easy integration with WhatsApp & websites	Restricted customization in free tier
LLM-based Chatbot (GPT / Mistral)	Transformer-based Generative AI	Multilingual (incl. Tamil & English)	93.4	0.68	High contextual accuracy, dynamic response generation	High computational cost, needs moderation
Proposed Hybrid NLP-LLM Chatbot (This Study)	Intent Recognition + LLM + Translation APIs	Tamil-English	95.1	0.49	Combines rule-based reliability with LLM creativity, multilingual SaaS-ready	Slightly higher setup complexity

The proposed SaaS Based Tamil and English Intelligent chatbot was evaluated using the RAGAS framework. The system uses intfloat/multilingual-e5-base to generate embeddings which is stored in ChromaDB for efficient semantic retrieval.

Instead of relying on traditional rule based methods such as pattern matching, Usage and integration of technologies such as Retrieval Augmented Generation (RAG), Natural Language Processing, Sentence transformers and IndicTrans Library enabled to process English and Tamil efficiently. Usage of these technologies has aided in overcoming difficulties faced during bilingual query handling faced previously.[9],[14],[16]

The system evaluated by means of RAGAS framework had context precision of 91.2%, faithfulness of 90% and an average response time of 1.2 seconds per query. Only a meagre 8% queries were escalated to human agents and 92% queries varying across Tamil and English languages were answered promptly and accurately by the chatbots generated. It also showed reduction in response latency of the SaaS Chatbot system through optimized embedding and usage of generative models.[13].

Therefore, the results of the proposed system indicate and demonstrate that the chatbot system is fast, dependable and aids in automating customer support sector, reducing workload and increasing turnover and customer retention proving its efficiency as a viable and economical tool for increasing sales by means of providing rapid and 24/7 available customer support.

VII CONCLUSION

The proposed SaaS based Chatbot platform with bilingual support for Tamil and English Support brings together the potential of Natural Language Processing, Generative AI and multilingual embedding in an attempt to automate the

customer support sector and help reduce human workload, widen the audience and increase business turnover.

It strays from the traditional rule based pattern matching and focuses on training chatbot model by means of efficient technologies such as sentence transformers and Retrieval Augmented Generation (RAG). The results of the system experiments prove and validate that these hybrid AI chatbots significantly enhance customer experience and efficiently resolve queries presenting them as scalable and viable solutions for increasing the brand reputation of Small and Medium sized Enterprises (SMEs).

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ABOUT THE CONFERENCE

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